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Designing multi-tier, multi-service-level, and multi-modal last-mile distribution networks for omni-channel operations

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ABSTRACT

The fast and dynamic evolution of e-retailing is driving structural changes in the way companies are reaching the urban consumer. Companies are offering faster delivery services, deploying collection-and-delivery points and multi-echelon distribution networks with a range of facility types, diversifying last-mile transportation modes with new types of vehicles and contracting third-party and crowd-sourcing-based transportation. In this paper, we propose an integrated modeling framework for strategic last-mile design of three-tiered multi-modal networks in omni-channel environment, with customer demand differentiated according to multiple time-differentiated delivery services and multiple product-exchange alternatives. In order to apply this framework in a large-scale demand setting and capture the granularity of customer demand and distribution network features, we extend the route cost estimation formulae from the literature and incorporate them in a three-echelon capacitated location-routing problem that optimizes the configuration of distribution networks in an integrated way. Through a number of numerical experiments, we show the impact of the diversified customer demand and the novel distribution network features on the overall network performance and demonstrate the economic benefit of an integrated approach. A real-world case study, inspired by the ongoing operations of a major omni-channel retailer in São Paulo, Brazil, serves to illustrate the real-world impact of our work.

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1. Introduction

While e-commerce represents approximately 10% of the global retail landscape, it is driving most of the growth in this sector. Projections suggest that e-retailing will continue to grow at an annual rate of approximately 20%, potentially turning it into a US$4 trillion market in 2020 (The Nielsen Company, 2017). While e-commerce accounted for only 14% of omni-channel sales in 2019 in the United States, it contributed 63% of the growth in omni-channel sales during the same period (Statista, 2020). E-retailing goes hand-in-hand with last-mile delivery services, including parcel and grocery deliveries. Overall, online shopping increases the fragmentation of deliveries to residential areas and to congested business districts, raising routing complexity and costs. In addition, the fast and dynamic evolution of omni-channel retailing is driving structural changes in the way companies are reaching the urban consumer.

To address rising customer expectations with respect to delivery lead time, location, and flexibility, e-retailers are repositioning their delivery service offerings. Omni-channel retailers see shorter delivery lead times as an opportunity to differentiate themselves from competitors (Wollenburg, 2017), leading numerous companies (e.g., JingDong (China) or Ulmart (Russia)) to promote expedited delivery services, such as same-day or instant delivery options (Janjevic & Winkenbach, 2020). Further, an omni-channel business-to-consumer (B2C) experience implies multiple physical and virtual interaction points with the customer (Brynjolfsson, Hu, & Rahman, 2013). Product exchange locations for online orders are therefore expanded beyond customers' homes or places of work, with e-retailers offering alternative delivery and returns options, such as pick-up and drop-off at collection and delivery points (CDPs).

The strategic design of the underlying last-mile distribution network must align with the delivery service offerings and the associated omni-channel strategy of the retailer (Janjevic & Winkenbach, 2020; Lim & Winkenbach, 2019). To support a growing set of differentiated delivery service offerings, companies are therefore required to adapt their existing distribution networks. To accommodate for the growing popularity of expedited delivery services, they are investing in multi-echelon distribution networks with diversified facility types and sizes. Multiple e-retailers (e.g., JingDong (China), B2W (Brazil)) and parcel operators servicing

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e-commerce deliveries (e.g., Chronopost (France), La Petite Reine (France)) are establishing hyper-local facilities, which often act as a third echelon of their distribution networks (Janjevic, Kaminsky, & Ballé Ndiaye, 2013; Janjevic & Winkenbach, 2020). Further, in light of new delivery requirements, increasing congestion and new environmental regulations, last-mile transportation modes are also evolving (Janjevic & Winkenbach, 2020). In addition to vans that have been a dominant mode of last-mile transportation in this sector, many companies are deploying cargo-bikes (e.g., UPS (global operations), La Petite Reine (France), three-wheelers (e.g., JingDong (China)) or motorcycles (e.g., Jumia (Nigeria), Flipkart (India)) (Janjevic & Winkenbach, 2020). Moreover, e-retailers increasingly rely on third-party operators to conduct their last-mile deliveries, including crowd-sourced transportation options (Simoni, Marcucci, Gatta, & Claudel, 2019).

The problem of finding an optimal last-mile network design in this context is of utmost relevance to the retail industry. However, logistics operations in omni-channel retailing have received limited attention in the academic literature, compared to the marketing or sales aspects of this retail strategy. Further, while a number of contributions addresses logistics operations in omni-channel settings (Cardenas, Sanchez-Diaz, and Dewulf, 2017; McLeod, Cherrett, and Song, 2006; Morganti, Dablanc, and Fortin, 2014; Weltevreden, 2008, see, e.g.,), very few of them propose design and planning models for distribution networks tailored to the specific requirements of omni-channel operations (Aksen and Altinkemer, 2008; Guerrero-Lorente, Ponce-Cueto, and Blanco, 2017; Janjevic, Merchán, and Winkenbach, 2019, see, e.g.,).

The extant literature on last-mile logistics modeling fails to appropriately address the challenge of designing omni-channel distribution networks for two main reasons. First, extant models fail to capture the *granularity* of state-of-the-art omni-channel distribution problems, which are characterized by *differentiated delivery service offerings* (i.e., multiple time-differentiated delivery services, alternative product exchange options), and *locally differentiated structural features* of the underlying last-mile distribution networks (i.e., multiple echelons, a range of facility types, sizes and roles, heterogeneous transportation modes). Some contributions address some of these features individually, such as the availability of crowd-shipping (Archetti, Savelsbergh, and Speranza, 2016; Archetti, Speranza, and Vigo, 2014; Arslan, Agatz, Kroon, and Zuidwijk, 2019; Kafle, Zou, and Lin, 2017, see, e.g.,) or same-day delivery option (Klapp, Erera, and Toriello, 2018; Voccia, Campbell, and Thomas, 2019, see, e.g.,), but focus on tactical and operational decisions rather than the strategic design of omni-channel distribution systems. Second, few works in the extant literature address the wide variety of strategic choices associated with the design of an omni-channel last-mile distribution network in an *integrated* yet computationally efficient approach. In other words, the extant literature does not yet provide a strategic network design model that simultaneously addresses all relevant aspects of the design problem, captures the non-trivial trade-offs between the various design decisions, and is scalable and transferable towards large-scale, real-world problem instances.

To close this apparent research gap, we introduce an *integrated modeling framework* for the strategic design of *three-tiered, multi-modal last-mile distribution networks* for omni-channel retailing, with customer demand being served via multiple *time-differentiated delivery services* and multiple *product-exchange options*. In order to apply this framework to *large-scale, real-world problem instances* and to capture the *granularity* of customer demand and distribution network features, we extend existing closed-form continuum approximations (CAs) of optimal routing cost from the literature to account for multiple product exchange options, transportation modes and route configurations. We incorporate these extended approximations in a three-echelon capacitated location-routing problem that optimizes the strategic design of the omni-channel distribution network in an integrated manner, i.e., by simultaneously determining the optimal number, type, and location of distribution facilities, the shape of facility-specific service areas, the modal choice, and the route configurations along which each individual delivery vehicle serves the underlying customer base.

Through a number of numerical experiments, we show the impact of explicitly capturing the *granularity* of omni-channel delivery service offerings and distribution network features during the network design stage on the overall network performance. Further, we demonstrate the benefit of an *integrated* optimization approach, as we are able to show that it proposes structurally different and better performing network designs than commonly employed non-integrated approaches. We then apply our model to a real-world case study, inspired by the ongoing operations of a major omni-channel retailer in São Paulo, Brazil, to illustrate the real-world applicability and impact of our work.

The remainder of this paper is structured as follows. In Section 2, we introduce the multi-tier, multi-service-level, and multi-modal last-mile network design problem. A review of the extant literature is presented in Section 3. Section 4 presents our proposed model formulation and solution method. In Section 5, we perform detailed numerical experiments in which we apply the proposed model to stylized problem instances. We then present and discuss the results from our real-world case study in Section 6. Section 7 presents our discussion of results and managerial insights. We conclude with a summary of our findings in Section 8.

2. Problem setting

We consider a strategic last-mile distribution network design problem for parcel-sized, non-perishable e-commerce deliveries by an omni-channel retailer. We first describe the characteristics of the customer demand, the delivery services offered, and the distribution network deployed. We then introduce the key components of the network design problem.

2.1. Customer demand characteristics and delivery service offerings

Large-scale demand E-commerce last-mile deliveries are characterized by a large-scale and highly fragmented demand base, comprising several thousand customers on a daily basis in a given city. For instance, the omni-channel retailer considered in our case study serves approximately 15,000 daily orders in the metropolitan area of São Paulo, Brazil. These deliveries are typically characterized by a small drop size (i.e., between 1 and 2 parcels per delivery).

Alternative product exchange options Depending on customer preferences, the delivery and product exchange can occur at customers' homes or at collection and delivery pointss (CDPs). CDPs can be implemented in various operating models and at different levels of automation. For example, our case study retailer decided to leverage its brick-and-mortar store network as attended pick-up and drop-off points, while other companies resort to fully automated parcel lockers. CDPs consolidate individual orders, giving rise to two types of points of demands (PODs) to be considered in our problem setting: (i) a large number of individual customers with small drop sizes, and (ii) relatively few CDPs with large, consolidated drop sizes.

Time-differentiated delivery services Customer demand is realized across multiple delivery services that differ according to their lead time. Specifically, we consider three increasingly common services offered in online retailing: standard delivery (e.g., next-day delivery for our case study company), express delivery (e.g., same-day

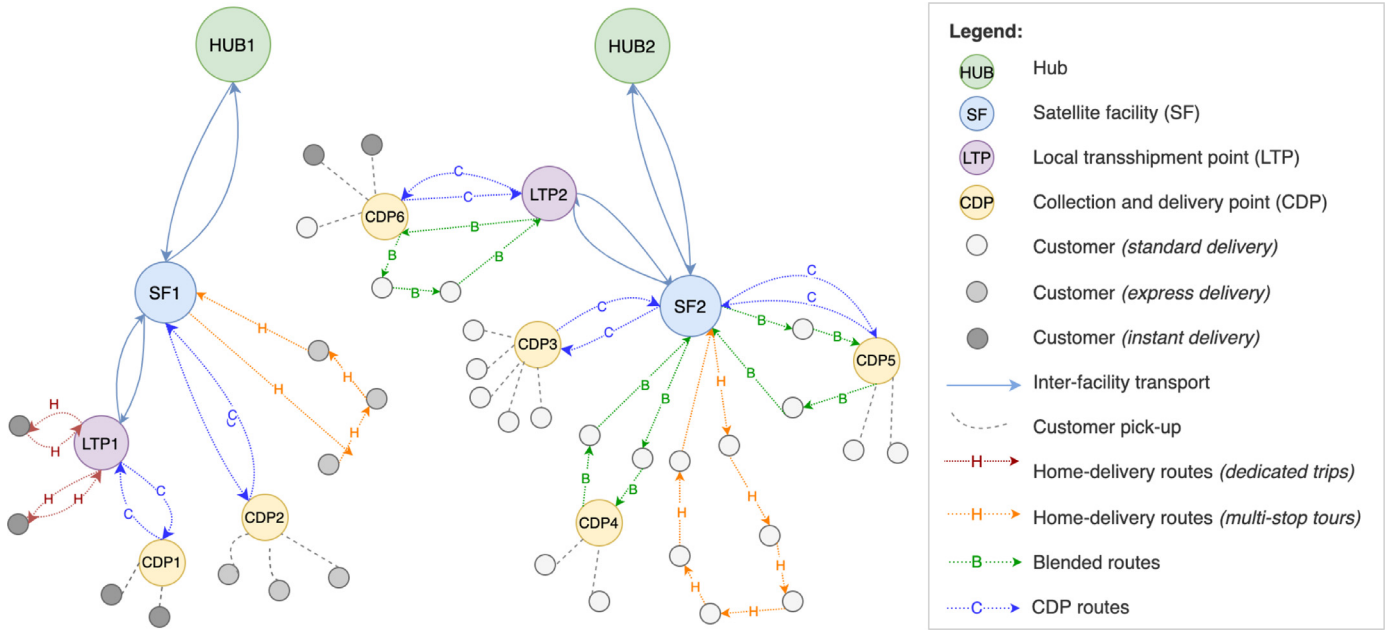


Fig. 1. Generalized illustration of the three-echelon distribution network architecture.

delivery for our company), and instant delivery (e.g., 4-hour deliveries for our company).

2.2. Distribution network characteristics

Fig. 1 illustrates the distribution network under investigation. Here, various types of nodes represent logistics facilities and product exchange locations. The arcs represent different route configurations deployed to move shipments in-between facilities and from facilities to PODs.

Three echelons of distribution facilities In response to the industry trends discussed in Section 1 and observed at our case study company, we consider three echelons of distribution facilities. *Hubs* constitute the first network echelon and serve as the origin of last-mile freight flows. These are generally large facilities where inventory is held, and parcels prepared after an online order has been placed. The second network echelon is formed by *satellite facilitiess (SFs)*, receiving shipments from hubs in large-capacity freight vehicles and allowing for temporary staging but not for permanent storage. At SFs, a fraction of parcels is sorted, consolidated and loaded onto last-mile delivery vehicles. The remaining parcels are transferred to *local transshipment points (LTPs)*. These are small, third-echelon facilities from where highly localized last-mile delivery routes are deployed. Consequently, two types of activities can take place at facilities: (1) *transfer of goods*, i.e., the (trans)shipment of parcels in order to (re-)route them towards other facilities (at hubs and SFs), and (2) *handling of goods*, i.e., the sorting, preparation, and loading of last-mile delivery routes that depart from a facility (at SFs and LTPs).

Different facilities can deploy last-mile distribution routes for specific delivery services. For example, in Fig. 1, we assume that *LTP1* serves only instant deliveries and that *SF1* serves only express deliveries. For each facility-delivery service pair, we define a global service time, i.e., a maximum time during which delivery routes need to be completed in order to meet the promised delivery lead time. This value is a function of the delivery lead time and the location of the facility in the network. For example, as expected, instant deliveries (e.g., 4-h delivery) will have a smaller global service time (e.g., 3 h) than express deliveries (e.g., same-day delivery with a maximum route duration of 6 h).

Logistics facilities at different echelons are connected through transportation links, consisting of inbound and outbound trips. These are established from hubs to SFs and from SFs to LTPs. Depending on the amount of freight flowing between facilities and the location of facilities (i.e., urban or suburban), different types of vehicles can be utilized to perform these trips.

Heterogeneous transportation modes and last-mile route configurations In order to fulfill customer orders, last-mile distribution routes are deployed from SFs and LTPs. These routes differ according to: (1) the transportation mode employed, and (2) the route configuration.

Last-mile transportation modes differ in terms of the physical and commercial characteristics of the vehicles used, and the governance of their operations. For instance, our case-study company envisions the deployment of several vehicle types (e.g., minivans, cargo-bikes) and can choose between company-owned, outsourced, and crowd-shipped governance schemes for these vehicles. The physical characteristics of the vehicles impact their operational features such as speed and capacity. The governance of vehicle operations impacts the commercial characteristics, i.e., the cost structure of these transportation modes. Company-owned vehicles typically result in considerable fixed cost for maintenance, insurance, depreciation, and so on. The exact cost structure of outsourced transportation is linked to the type of subcontracting and the compensation scheme. For example, if entire vehicle routes are outsourced, subcontractors can charge a price per unit of distance or time. If individual orders are outsourced, a compensation scheme per parcel can be deployed. Different transportation modes and governance schemes are generally more relevant to specific delivery services. In the example in Fig. 1, we assume that standard and express deliveries are served thought the company's own fleet of minivans, while instant deliveries are serviced through crowd-shipping.

Last-mile route configuration differs according to (1) the type of PODs served (i.e., customers, CDPs, or both), and (2) the level of consolidation, i.e., the number of PODs per route.

We distinguish three types of routes according to the types of PODs served: *home delivery routes*, servicing customer homes only, *CDP routes*, servicing CDPs only, and *blended routes*, servicing customer homes and CDPs jointly. For *home delivery routes*, we

further distinguish between two types of routes according to the level of consolidation: *dedicated trips*, where a single customer is served per route, and *multi-stop tours*, where several PODs are served per route. In Fig. 1, instant deliveries are served through *dedicated trips* due to their shorter lead times, whereas standard and express deliveries allow for more consolidation and are thus served through *multi-stop tours*.

If CDPs are present in the service area, the type of routes deployed first depends on the type of route configuration for home deliveries. If *dedicated trips* are performed for home deliveries, CDPs are served through *CDP routes* only (see, e.g., routes departing from *LTP1* in Fig. 1). If *multi-stop tours* are used for home deliveries, we make a second distinction regarding the joint deliveries towards customer homes and CDPs. In a first case, the different operational requirements of these PODs result in separate routes towards customer homes and CDPs. Consequently, *home delivery routes* are organized through *multi-stop tours* while (dedicated) *CDP routes* are used for delivering to CDPs. For instance, in Fig. 1, express deliveries towards customer homes and CDPs are serviced through separate routes (see, e.g., routes departing from *SF1*). In a second case, CDPs and home deliveries are served jointly. In Fig. 1, we assume that standard deliveries allow for joint servicing of CDPs and home deliveries. In this case, the types of routes deployed depend on the volume of parcels at the CDPs. Finally, if one CDP accommodates multiple delivery services, it can be be served by a combination of *CDP routes* and *blended routes* (see, e.g., routes servicing *CDP6* in Fig. 1).

2.3. Strategic last-mile network design problem

We consider that demand characteristics, such as the location of different PODs and the delivery service requirements in terms of speed and product exchange options, are known. Therefore, the strategic design problem aims at establishing a last-mile distribution network that yields an optimal cost performance, while meeting the diversified customer requirements faced by the omni-channel retailer.

Design objectives Our aim is to find a distribution network design that minimizes the overall cost of distribution. We consider cost associated to the establishment and operation of logistics facilities (i.e., fixed facility costs and handling costs) as well as to the deployment of transportation routes linking different parts of the distribution system (i.e., inter-facility and last-mile distribution costs). Considering that the individual demands for the different delivery services offered are assumed to be given, revenues generated by different services or incentives to promote one service over the other (e.g., incentives for delaying a delivery or using a CDP) are outside of the problem scope considered here.

Design decisions The design and planning of an urban last-mile distribution network involves three levels of decisions: (i) long-term strategic decisions (e.g., network architecture and facility locations), (ii) medium-term tactical decisions (e.g., size and composition of the vehicle fleet); and (iii) short-term operational decisions (e.g., vehicle routing) (Bektas, Crainic, & Van Woensel, 2016; Crainic & Laporte, 1997; Min, Jayaraman, & Srivastava, 1998). The strategic, multi-tier, multi-modal network design problem considered in this work gives rise to several sets of design decisions that define different components of the distribution network. The first set of decisions defines the network architecture along the three levels of *logistics facilities* previously discussed (i.e., hubs, SFs and LTPs). It consists in determining the optimal number, type, and location of these facilities. The second set of decisions defines the *inter-facility transportation* flows required to operate the network. For each pair of facilities in the network, it is necessary to determine the number of parcels shipped from one facility to the other, as well as the size and composition of the required vehicle fleet. The third set of decisions defines the *allocation of customer demand*. First, all PODs, each characterized by a preferred delivery service (i.e., standard, express, or instant delivery) and product exchange option (i.e., home delivery or CDP), need be allocated to a serving facility. Second, the transportation modes (i.e., different vehicle types and governance schemes) serving these PODs need to be chosen. The fourth set of decisions defines the configuration of the *last-mile delivery routes*. These decisions include determining the required fleet size and the optimal sequence in which PODs are served along the last-mile delivery routes. While these operational decisions play a sub-servant role in strategic network design, they cannot be entirely ignored given their strong interdependecies with other design decisions. These are detailed in the next section.

Interdependencies between design decisions The aforementioned design decisions exhibit strong interdependencies, which gives rise to the need to consider them simultaneously in an integrated optimization approach, to account for the non-trivial cost trade-offs that emerge from these interdependencies. For example, increasing the number of logistics facilities and their proximity to customers leads to an increase in fixed facility costs, but also to a reduction in last-mile transportation cost. While this trade-off exists in numerous operational settings, it is even more critical for time-constrained last-mile operations. On the one hand, locating facilities closer to demand decreases the cost of dedicated trips. On the other hand, decreased line-haul travel enables a higher level of route consolidation under lead-time constraints, and can lead to significant efficiency gains. The industry examples discussed in Section 1 illustrate how companies progressively implement decentralized network architectures with a higher number of second- and third-echelon facilities, in order to distribute goods more efficiently as expedited delivery services are becoming increasingly popular.

Further, last-mile transportation mode choices are strongly dependent on the location and demand coverage of facilities. For instance, facilities located in high-demand areas may enable the efficient use of cargo-bikes or pedestrian deliveries. In addition, given their more flexible compensation scheme, outsourced or crowdsourced deliveries may constitute an attractive option for expedited delivery services. The ability to outsource deliveries to some areas, in turn, has repercussions on the overall network configuration, as it reduces the need for dedicated facilities in low-density areas.

Finally, the introduction of CDPs can affect the demand patterns in a service area given their ability to consolidate demand. This impacts the structure of the delivery routes in their surroundings, which can in turn affect the optimal design of the upstream network.

All in all, as service offerings in terms of delivery speed and mode of product exchange continue to diversify, the aforementioned design decisions become increasingly interdependent and the need to address the complex design trade-offs resulting from these interdependencies in an integrated optimization approach becomes increasingly relevant.

3. Literature review

In the following, we first provide an overview of the extant literature on modeling the strategic design of multi-echelon distribution networks in last-mile logistics in Section 3.1. We identify the dominant modeling approach in the extant literature and the corresponding class of optimization models, referred to as multi-echelon location-routing problems (ME-LRPs). In Section 3.2, we then review the extant academic contributions related to these models to provide insights into their ability to capture the relevant features of the design problem outlined in Section 2.3.

3.1. Multi-echelon network design in last-mile logistics

Network design models for multi-echelon distributions systems have been widely studied in the literature (see, e.g., Laporte, 1988). Nevertheless, specific applications to urban last-mile logistics systems have only been reported over the past 15–20 years. One of the earliest contributions in this field is Taniguchi, Noritake, Yamada, and Izumitani (1999), who propose a bi-level optimization model building on queuing theory and non-linear optimization techniques to determine the size and location of multi-company (public) consolidation terminals. Subsequent contributions in this field generally follow two different modeling approaches, corresponding to two classes of optimization problems: location-allocation problems (LAPs) and location-routing problems (LRPs).

LAPs abstract from any tactical and operational decisions related to the network's fleet of vehicles and their routing. The cost to serve each POD is typically modeled using straight-line or radial distances (Chien, 1993). We find multiple contributions applying LAPs in the context of multi-echelon last-mile distribution network design (see e.g., Baldi, Ghirardi, Perboli, & Tadei, 2012; Charisis & Kaisar, 2019; Crainic, Ricciardi, & Storchi, 2004). Further, a handful of contributions specifically consider operations in an omni-channel environment (see e.g., Guerrero-Lorente et al., 2017; Min, Ko, & Ko, 2006).

However, as discussed in Section 2.3, location and routing decisions are highly interrelated. Using distance between a facility and PODs as a proxy for routing costs fails to capture the characteristics of last-mile distribution routes that typically consolidate multiple orders. Multiple authors find that solving location and routing problems independently results in sub-optimal solutions (Balakrishnan, Ward, & Wong, 1987; Bruns, 1998; Nagy & Salhi, 2007; Salhi & Rand, 1989; Tuzun & Burke, 1999; Webb, 1968). Integrating strategic facility location decisions with operational routing decisions gives rise to the problem class of integrated LRPs and their multi-echelon counterparts, ME-LRPs. The vast majority of recent contributions in the field of strategic last-mile multi-echelon network design adopt this approach. A review of the most relevant contributions in the LRP and ME-LRP space is provided in the following section.

3.2. Multi-echelon location-routing problems

The literature on LRPs is fairly extensive and predominantly focused on single-echelon distribution systems (see e.g., Cuda, Guastaroba, & Speranza, 2015; Drexl & Schneider, 2015; Nagy & Salhi, 2007; Prodhon & Prins, 2014; Schneider & Löffler, 2019 for a review). More recently, the academic literature has focused on ME-LRPs, which extend the concept of integrated LRPs to the design of multi-echelon distribution networks. For example, a number of papers explore location-routing extensions for the two-echelon city logistics distribution systems described in Crainic et al. (2004), (Boccia, Crainic, Sforza, and Sterle, 2010, 2011; Contardo, Hemmelmayr, and Crainic, 2012; Crainic, Sforza, and Sterle, 2011, see, e.g.,).

Table 1 presents a comprehensive summary of the relevant ME-LRPs literature. Here, we map the surveyed contributions to the critical features of the problem setting addressed in this paper, as discussed in Section 2. Further, we distinguish the reviewed contributions according to the methodological approach they employ. We find two broad families of methods used to incorporate the routing component in the overarching ME-LRP: (1) *discrete methods*, and (2) *methods based on continuum approximation (CA) approaches*. Discrete methods explicitly consider vehicle routing decisions to serve each individual POD along an optimum or near-optimum sequence of stops. In contrast, methods employing CA approaches deliberately ignore detailed distance calculations between PODs and do not seek an exact solution for the optimal stop sequence of any particular vehicle route (Ansari, Başdere, Li, Ouyang, & Smilowitz, 2018). Instead, they aggregate demand data in a given area, use routing cost estimations (RCEs) to approximate near-optimal route distances and cost within the area and incorporate them into a higher-level ME-LRP. By using RCEs, the ME-LRP is essentially reduced to a multi-echelon LAP. However, in contrast with classical LAPs, RCEs employ a much more granular routing cost expression that includes detailed cost estimates for each relevant component, namely line-haul travel time and distance, local travel times and distances, and service times at PODs. They therefore approximate routing decisions and their dependency on location decisions sufficiently well, while allowing a substantial reduction in computation times (Nagy & Salhi, 2007) compared to LRPs based on discrete methods. RCEs methods are particularly well suited for large-scale applications, where routing decisions play a subordinate role at the strategic design stage and demand variations across its different realizations (e.g., different customers to be served each day) limit the usefulness of explicit routing decisions. In the following, we review the extant literature on ME-LRPs employing discrete and CA-based approaches along the critical features of the problem setting described in Section 2.

Large scale demand setting As shown in Table 1, discrete methods fail to address problem instances of realistic size in an e-commerce / omni-channel retailing context with several thousands of customers. Many models aiming to find an optimal solution to the ME-LRP using exact methods become intractable even for relatively small problem instances (e.g., Boccia, Crainic, Sforza, & Sterle, 2011 consider problems instances with 25 customers, while Ambrosini, Routhier, Toilier, and others (2005) consider problem instances with 135 customers). Therefore, a large body of the ME-LRP literature focuses on heuristic and metaheuristic approaches, which allow to increase the problem size, typically at the expense of a loss in optimality. For example, Boccia et al. (2010), Contardo et al. (2012), Schwengerer, Pirkwieser, and Raidl (2012), Schneider and Löffler (2019) and Nguyen, Prins, and Prodhon (2012) develop solution methods for the 2E-LRP formulated by Boccia et al. (2011). Given their similarity with regards to the problem considered, we do not include them in Table 1. Still, real-world, large-scale applications of ME-LRPs are computationally infeasible even for *state-of-the-art* metaheuristics. To the best of our knowledge, the largest problem instance addressed by any extant ME-LRP is presented by Nikbakhsh and Zegordi (2010), who consider instances with up to 500 customers.

The integration of RCE methods into the routing component of the LRPs leads to drastic reductions in the required computational effort, which enables the consideration of larger-scale problem instances. Given that demand information is spatially aggregated, there is strictly speaking no limitation to the number of PODs that can be considered. For example, Merchán, Winkenbach and Snoeck (2020) consider 16,000 PODs and Winkenbach, Kleindorfer, and Spinler (2016a) consider up to 149,000 PODs in a two-echelon location-routing problem (2E-LRP).

Time-differentiated delivery services We consider multiple delivery services that differ according to their delivery lead time. The temporal aspects of deliveries can be incorporated in discrete modelling approaches by introducing customer-specific time-windows and maximum service or driving times. For instance, Nikbakhsh and Zegordi (2010) and Govindan, Jafarian, Khodaverdi, and Devika (2014) study the 2E-LRPs with time windows. The literature on ME-LRPs using CA-based methods offers several contributions in which maximum service time constraints are incorporated (Janjevic et al., 2019; Merchán and Winkenbach, 2017; Snoeck and Winkenbach, 2020; Winkenbach et al., 2016a; Winkenbach, Roset, and Spinler, 2016b, see, e.g.,). We note, however, that these contributions consider a single delivery service with no time differentiation (i.e., a unique maximum service time). Since the academic lit-

Table 1
Selected references on. ME-LRP.

Reference	Customer demand and delivery services			Distribution network	
	Large scale demand setting	Time-differentiated delivery services	Alternative product exchange points	More than two echelons	Heterogeneous transp. modes and routes
Discrete methods					
Aksen and Altinkemer (2008)			x		
Ambrosino and Grazia Scutellà (2005)				x	
Ambrosino et al. (2009)				x	
Boccia et al. (2011)					
Hamidi et al. (2012)				x	
Hamidi et al. (2014)			x		
Lee et al. (2010)				x	
Lin and Lei (2009)			x		
Nikbakhsh and Zegordi (2010)		x			
Govindan et al. (2014)		x			
CA-based methods					
Guerrero-Lorente et al. (2020)	x	x	x		
Merchán and Winkenbach (2017)	x				x
Jahangiriesmaili, Bahrami, and Roorda (2017)	x				
Janjevic et al. (2019)	x		x		
Snoeck and Winkenbach (2020)	x				x
Winkenbach et al. (2016a)	x				x
Present study	x	x	x	x	x

erature on CA approaches that take into account customer-specific time windows is still scarce (see Figliozzi (2009) for one of the few contributions in this direction), such time window constraints have not yet been integrated in CA-based LRPs available in the extant literature.

Alternative product exchange options Only a handful of contributions explicitly consider deliveries towards alternative product exchange points (i.e., individual customer homes and CDPs). Aksen and Altinkemer (2008) consider a two-echelon network composed of warehouses and brick-and-mortar stores. The stores are used both as origin of last-mile delivery routes serving online customers and delivery pick-up points. Note that alternative product exchange points can be incorporated into discrete methods by modeling CDPs as customer locations with a larger demand and longer service times. Although they do not explicitly consider CDPs, such an approach is used by Lin and Lei (2009) who consider two types of customers with significantly different demand characteristics. Guerrero-Lorente, Gabor, and Ponce-Cueto (2020) and Janjevic et al. (2019) both consider deliveries towards individual customers and CDPs and employ CA-based methods. Here, Guerrero-Lorente et al. (2020) aggregate demand data across different PODs and then apply the augmented routing cost estimations (ARCEs) formulae proposed by Winkenbach et al. (2016a). However, as discussed in Section 2.2, the presence of CDPs leads to the emergence of multiple possible configurations of last-mile delivery routes with different characteristics. To account for these additional complexities, Janjevic et al. (2019) introduce an extended ARCE formulation based on Winkenbach et al. (2016a). They benchmark their solution against a set of academic instances and show that this approach systematically outperforms the original formulation in a setting with heterogeneous PODs.

More than two echelons of distribution Several extant contributions consider more than two echelons of logistics facilities in a ME-LRP (Ambrosino and Grazia Scutellà, 2005; Hamidi, Farahmand, Reza Sajjadi, and Nygard, 2012; Hamidi, Farahmand, Sajjadi, and Nygard, 2014; Lee, Moon, and Park, 2010, see, e.g.,). These works all adopt a discrete-modeling approach for limited-size problem instances. Moreover, none of these contributions consider a last-mile distribution setting.

Heterogeneous modes and configurations of last-mile delivery routes We consider multiple vehicle types and multiple types of fleet governance and route configurations. Examples for ME-LRP with heterogeneous fleets can be found in the literature for both discrete (Ambrosino, Sciomachen, and Scutellà, 2009, see, e.g.,) and CA-based approaches (Merchán and Winkenbach, 2017; Snoeck and Winkenbach, 2020; Winkenbach et al., 2016a, see, e.g.,). Heterogeneous vehicle fleets also allow to model various governance options. For example, outsourcing entire vehicle routes can be modelled by considering vehicle options that reflect an outsourcing scheme in their cost parameters. To the best of our knowledge, outsourcing of individual orders through the use of crowd-shipping schemes has only been explored in the extant literature at the level of tactical and operational decisions (Archetti et al., 2016; Archetti et al., 2014; Arslan et al., 2019; Kafle et al., 2017, see, e.g.,).

3.3. Literature gap

Table 1 shows that none of the reviewed contributions provides a strategic network design model that comprehensively addresses all aspects of our problem setting. ME-LRPs employing discrete methods allow to account for more granular characteristics of demand and delivery services but fail to address large-scale settings. CA-based approaches, on the other hand, allow for applications to large-scale problem instances but typically fail to capture the granular nature of omni-channel / e-commerce demand. Further, extant literature hardly considers recently emerging features of last-mile distribution networks that are specifically tailored to serve e-commerce demand in an omni-channel environment, such as three-echelon distribution network configurations and new types of transportation governance options and route configurations.

Our study addresses this gap through the following methodological contributions: (1) elaboration of an optimization model for a three-echelon capacitated location-routing problem, (2) extension of the augmented RCE formulae from the literature to account for the granular description of customer demand, new types of transportation governance and route organization.

4. Methodology

We formulate an optimization model for a three-echelon capacitated location-routing problem, incorporating multiple delivery services, product exchange options, and last-mile transportation

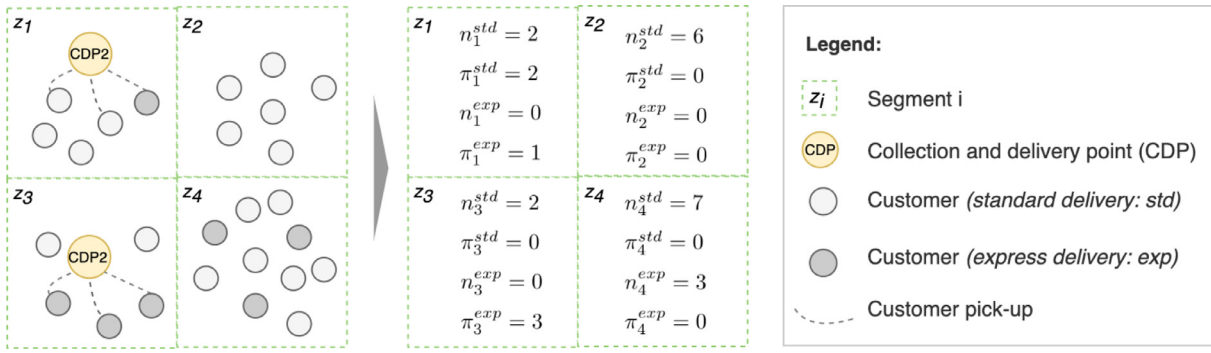


Fig. 2. Aggregation of customer demand.

modes. To apply this model to a *large-scale demand setting*, we build on a CA-based method introduced by Winkenbach et al. (2016a), in which an augmented RCE formula is used to approximate routing cost within an LRP. Further, to account for non-homogeneous spatial distributions of demand, Winkenbach et al. (2016a) divide the service area into a dense grid of quadratic segments which are then characterized according to parameters such as average demand density or drop-size. This raster data model approach to discretize the service area is consistent with the underlying assumptions of CA-based route approximations: quadratic segments ensure a convex shape of the service sub-area, while their large number grants them relatively small surface areas, allowing to model the demand intensity within each segment as a two-dimensional homogeneous point process.

We extend this approach by considering a more granular description of the customer demand per segment, differentiating it according to several delivery services offered. Fig. 2 illustrates this approach. Here, the service area is subdivided in four segments z_i, with $i \in \{1, 2, 3, 4\}$. Each segment z_i is characterized by an aggregate demand parameter for each type of delivery service. n_i^{std} and n_i^{exp} represent the number of home delivery customers for standard and express deliveries. π_i^{std} and π_i^{exp} represent the CDP demand for standard and express deliveries.

Further, to account for two different types of PODs (i.e., home deliveries and CDP deliveries), we rely on a method introduced by Janjevic et al. (2019) that extends the RCE of Winkenbach et al. (2016a) to account for deliveries towards heterogeneous demand points. Finally, to account for multiple types of transportation modes that differ in terms of vehicle type, fleet governance, and route configuration, we further extend the RCE formulations of Winkenbach et al. (2016a) and Janjevic et al. (2019) to enable full flexibility to capture arbitrary vehicle cost structures. In summary, our methodological contribution entails extensions to the extant RCE formulations to account for differentiated delivery services and alternative transportation modes. These extensions enable us to solve the ME-LRP for a three-echelon distribution network with multiple logistics facility types, product-exchange options, delivery service types, and route configurations, which has not yet been addressed in the extant literature.

The remainder of this Section is structured as follows. In Section 4.1, we formally define our distribution network. We then formulate an optimization model to inform the strategic design of this network in Section 4.2. The overall model formulation is provided in Section 4.2.1, while Section 4.2.2 presents the extended RCE formulae used for the routing component of our model.

4.1. Definition of the distribution network

In this section, we formally describe the various components of the distribution network, as presented in Section 2.2. The notation and definitions of all relevant model parameters are listed in Table 2.

4.1.1. Customer demand and delivery services

We consider customer demand in a service area divided into a set of rectangular segments Z, of area A_z. Multiple delivery services, differentiated by their delivery lead-time, can be deployed. Let X be the set of available delivery services. For each combination of delivery service $x \in X$ and segment $z \in Z$, we define the number of home delivery stops, n_z^x, the average drop-size for home deliveries, ρ_z^x, and the number of CDP parcels, π_z^x. We also define the average physical volume per parcel for both home and CDP deliveries, θ^x, which applies to demand across all segments.

4.1.2. Logistics facility types and inter-facility transportation

Let F be a finite and discrete set of candidate facilities with $H \subseteq F$, $S \subseteq F$, and $L \subseteq F$ representing subsets of candidate facilities for hubs, SFs and LTPs, respectively, and $F = \{H \cup S \cup L\}$. For every facility $f \in F$, the parameter t_f represents the network tier associated to the facility (i.e., 1 for hubs, 2 for SFs and 3 for LTPs) and Ω_f denotes its maximum throughput capacity. Each facility is characterized by cost parameters c_f^{F}, c_f^{T} and c_f^{H}, representing its fixed cost, per-parcel transfer cost, and per-parcel handling cost, respectively.

For each combination of facility $f \in F$ and segment $z \in Z$, parameter r_{fz} represents the travel distance between f and the centroid of z. To model constraints on which delivery service can be provided from a given facility, the parameter parameter a_{fx} takes value 1 to indicate that service type $x \in X$ can be provided from facility $f \in F$ and zero otherwise. Depending on the maximum lead time of the delivery service $x \in X$ and the time required to prepare orders and potentially transfer them to a facility $f \in F$, different cut-off times for the deployment of last-mile delivery routes are used. We therefore define per delivery service $x \in X$ and facility $f \in F$, a global service time T_{fx}^{m}, representing the maximum time during which delivery routes can be deployed.

Inter-facility transportation links can be established between any two facilities $f \in F$ and $g \in F$ that are associated to consecutive network tiers, i.e., if $t_g = t_f + 1$. The parameter d_{fg} denotes the distance between the facility $f \in F$ and $g \in F$. V represents the set of vehicle types that can perform inter-facility transportation. For every $v \in V$, we define c_v^{km} and ξ_v, denoting the distance-based operational cost and vehicle capacity, respectively. We assume that there are no limitations with regards to the size of the vehicle fleet deployed.

4.1.3. Last-mile distribution routes

Last-mile transportation modes Let T be the set of transportation modes that can be deployed for last-mile delivery routes. For every $t \in T$, we define cost parameters c_t^{km}, c_t^{H}, c_t^{p}, and c_t^{F}, denoting

Table 2
General model parameters.

Symbol	Description	Unit
General model parameters		
t_f	tier level of the facility $f \in F$	
d_{fg}	travel distance between the facility $f \in F$ and the facility $g \in F$	[km]
r_{fz}	travel distance between facility $f \in F$ and the center of the demand zone segment $z \in Z$	[km]
A_z	area of demand zone segment $z \in Z$	[km^2]
κ_z	circuity factor in the demand zone segment $z \in Z$	
κ_t	transportation mode-specific detour factor on the second echelon for $t \in T$	
a_{xf}	parameter describing if facility $f \in F$ can serve delivery service $x \in X$	
b_{xt}	parameter describing if last-mile transportation mode $t \in T$ can serve delivery service $x \in X$	
T^{m}_{fx}	maximum global service time for delivery service $x \in X$ when deployed from facility $f \in F$	[hour]
λ^x	parameter describing route configuration of *home delivery routes* for delivery service $x \in X$	
τ^x	parameter describing route consolidation for delivery service $x \in X$	
Demand parameters		
n^x_z	number of home delivery stops in demand zone segment z for delivery service x	[customers]
ρ^x	drop size of parcels for individual customers for delivery service x	[parcels/customer]
π^x_z	number of parcels at CDPs for delivery service x	[parcels/CDP]
θ^x	average physical volume per parcel for delivery service x	[m^3/parcel]
Operational parameters		
Ω_f	maximum capacity of a facility $f \in F$	[parcels]
t^{l}_t	time for vehicle loading and tour setup for transportation mode $t \in T$	[hour]
t^{d}_t	fixed time per delivery for transportation mode $t \in T$	[hour]
t^{i}_t	incremental time per delivery at POD for transportation mode $t \in T$	[hour/parcel]
ξ_v	volumetric carrying capacity of vehicle type $v \in V$	[m^3]
ξ_t	volumetric carrying capacity of transportation mode $t \in T$	[m^3]
s_v	average inter-facility transport speed for vehicle $v \in V$	[km/hour]
s^{l}_t	average line-haul speed for transportation mode $t \in T$	[km/hour]
s^{s}_t	average inter-stop speed for for transportation mode $t \in T$	[km/hour]
Cost parameters		
c^{F}_f	fixed cost to enable a facility in location $f \in F$	[$]
c^{T}_f	per-parcel transfer cost at facility $f \in F$	[$/ parcel]
c^{H}_f	per-parcel handling cost for last-mile distribution at facility $f \in F$ for delivery service $x \in X$	[$/ parcel]
c^{km}_v	operational distance-based cost for vehicle type $v \in V$	[$/km]
c^{km}_t	operational distance-based cost for transportation mode $t \in T$	[$/km]
c^{H}_t	operational time-based cost for transportation mode $t \in T$	[$/hour]
c^{F}_t	daily fixed vehicle cost for transportation mode $t \in T$	[$/day]
c^{p}_t	per-parcel cost for transportation mode $t \in T$	[$/parcel]

the distance-based operating cost, time-based operating cost, per-parcel delivery cost and daily fixed vehicle cost of vehicles, respectively. Depending on the governance of the transportation modes and the corresponding compensation schemes, different cost parameters are relevant and some can be null. For example, for company-owned vehicles, only c^{km}_t, c^{H}_t, and c^{F}_t will be considered. For crowd-shipping transportation modes, a compensation scheme per parcel and per distance (i.e., non-zero parameter values for c^{p}_t, c^{km}_t) could be put in place.

Each $t \in T$ is further characterized by parameters ξ_t, s^{l}_t, and s^{s}_t, denoting the volumetric carrying capacity, the average line-haul speed and the average inter-stop speed, respectively. Parameter t^{l}_t expresses the time for vehicle loading and tour setup for transportation mode $t \in T$. The stop duration is composed of a fixed duration per delivery, t^{d}_t, and an incremental stop duration per parcel delivered, t^{i}_t. Finally, for each last-mile transportation mode $t \in T$ and delivery service $x \in X$, parameter b_{xt} takes value 1 to indicate that service x can be provided on mode t and zero otherwise. We assume that there are no limitations with regards to the mix of transportation modes employed or the size of the corresponding vehicle fleets.

To capture the effect of road network features on travel directness for different transportation modes, we use two parameters. First, each segment $z \in Z$ is characterized by its circuity factor κ_z. Circuity is usually measured as the expected ratio between the true network distance and the Euclidean distance between any two points in the segment (see, e.g., Merchán and Winkenbach, 2019, for a detailed discussion of road network circuity). Further, each last-mile transportation mode $t \in T$ is characterized by a specific detour factor κ_t. This parameter allows us to adjust the average circuity within each segment, κ_z, to express differences in travel directness between different types of vehicles, which may result from differences in vehicle size and applicability of traffic regulations.

Last-mile route configuration Let $R = \{c, h, b\}$ be the set of delivery route types that can be deployed, namely *home delivery routes*, *CDP routes*, and *blended routes* (see Section 2.2). Further, parameter λ^x describes the route configuration for *home delivery routes* for delivery service $x \in X$ and takes value 1 if *multi-stop tours* are used and zero if *dedicated trips are used*. Finally, in case of *multi-stop tours*, the parameter τ^x can take a value 1 if blending home and CDP deliveries is allowed, and zero otherwise.

4.2. Optimization model

The optimization model proposed in the following addresses the integrated last-mile distribution network design problem as described in Section 2 and the corresponding network design decisions.

4.2.1. Model formulation

The overview of decision variables is given in Table 3. The model determines the optimal number and locations of *logistics facilities* from a given set of candidate nodes using a variable y_f. To decide on *inter-facility transportation* flows, the model allocates LTPs to SFs and SFs to hubs with a continuous decision variable ϕ^x_{fgv} varies and the integer variable ψ^x_{fgv} to determine the optimal number of vehicles needed to accommodate these flows. Lastly, the

Table 3
Decision variables.

Symbol	Description
y_f	binary variable indicating if a facility location $f \in F$ is active
ϕ^x_{fgv}	continuous variable representing the number of parcels relevant to the delivery service x between facility f and facility g with vehicle type v
ψ^x_{fgv}	integer variable representing the number of vehicles relevant to the delivery service x between facility f and facility g with vehicle type v
x^x_{zft}	binary variable denoting whether or not a demand for the delivery service x in a demand zone segment z is served from the facility f with last-mile transportation mode t

model *allocates customer demand* from each segment for each delivery services to an active facility and available last-mile transportation mode with a binary decision variable x^x_{zft}.

The objective of the model is to minimize the overall cost of distribution,

$$\min \left(K^{\mathrm{F}} + K^{\mathrm{T}} + K^{\mathrm{l}} + K^{\mathrm{H}} + K^{\mathrm{l}}\right), \tag{1}$$

where

$$K^{\mathrm{F}} = \sum_{f \in F} y_f c^{\mathrm{F}}_f, \tag{2}$$

$$K^{\mathrm{T}} = \sum_{f \in F} \sum_{g \in F} \sum_{v \in V} \sum_{x \in X} c^{\mathrm{T}}_f \phi^x_{fgv}, \tag{3}$$

$$K^{\mathrm{l}} = \sum_{f \in F} \sum_{g \in F} \sum_{v \in V} \sum_{x \in X} 2 c^{\mathrm{km}}_v d_{fg} \psi^x_{fgv}, \tag{4}$$

$$K^{\mathrm{H}} = \sum_{f \in F} \sum_{x \in X} \sum_{z \in Z} \sum_{t \in T} c^{\mathrm{H}}_f x^x_{zft} (n^x_z \rho^x_z + \pi^x_z), \tag{5}$$

$$K^{\mathrm{l}} = \sum_{x \in X} \sum_{t \in T} \sum_{f \in F} \sum_{z \in Z} x^x_{zft} C^x_{zft}, \tag{6}$$

subject to

$$\sum_{f \in F} \sum_{t \in T} x^x_{zft} = 1, \qquad \forall z \in Z, x \in X, \tag{7}$$

$$x^x_{zft} \le y_f a_{ft} b_{xt}, \qquad \forall z \in Z, x \in X, t \in T, f \in F, \tag{8}$$

$$\phi^x_{fgv} \le M(1 - \mid t_g - t_f - 1 \mid), \qquad \forall f \in F, g \in F, x \in X, v \in V, \tag{9}$$

$$\frac{\phi^x_{fgv}}{\xi_v} \le \psi^x_{fgv}, \qquad \forall f \in F, g \in F, x \in X, v \in V, \tag{10}$$

$$y_f \Omega_f \ge \sum_{g \in F} \sum_{x \in X} \sum_{v \in V} \phi^x_{fgv} + \sum_{x \in X} \sum_{z \in Z} \sum_{t \in T} x^x_{zft} (n^x_z \rho^x_z + \pi^x_z) \qquad f \in F, \tag{11}$$

$$\sum_{g \in F} \sum_{v \in V} \phi^x_{gft} = \sum_{g \in F} \sum_{v \in V} \phi^x_{fgv} + \sum_{z \in Z} \sum_{t \in T} x^x_{zft} (n^x_z \rho^x_z + \pi^x_z) \qquad f \in S \cup L, x \in X, \tag{12}$$

$$y_f \in \{0, 1\}, \qquad \forall f \in F, \tag{13}$$

$$x^x_{zft} \in \{0, 1\}, \qquad \forall z \in Z, f \in F, x \in X, t \in T, \tag{14}$$

$$\phi^x_{fgv} \ge 0, \qquad \forall f \in F, g \in F, x \in X, v \in V, \tag{15}$$

$$\psi^x_{fgv} \in \mathbb{N}, \qquad \forall f \in F, g \in F, x \in X, v \in V. \tag{16}$$

Eq. (2) provides the fixed cost of activated facilities, K^{F}. For each facility $f \in F$, this term is computed by multiplying the value of y_f with parameter c^{F}_f. Eq. (3) provides the inter-facility transfer cost. At each facility $f \in F$, this cost is obtained by multiplying the per-parcel transfer cost c^{T}_f by the number of transferred parcels, ϕ^x_{fgv}. Eq. (4) provides the inter-facility transportation cost K^{l}. For each pair of facilities $f \in F$ and $g \in F$, it is computed considering the round-trip distance between the facilities, d_{fg}, the number of vehicles used, ψ^x_{fgv}, and the distance-based operating cost of the corresponding vehicles, c^{km}_v. Eq. (5) provides the handling cost for last-mile distribution routes K^{H}. At each facility f, this value is computed by multiplying the per-parcel handling cost, c^{H}_f, by the number of parcels distributed from f. Eq. (6) provides the cost of last-mile distribution K^{l}, where C^x_{zft} is the routing cost associated with providing delivery service x to segment z from facility f through transportation mode t. Its computation relies on the extended CA-based RCE formulae we are proposing in Section 4.2.2 below.

Constraints (7) signify that demand in every segment $z \in Z$ and for every delivery service $x \in X$ must be allocated to exactly one facility f and transportation mode t. Constraints (8) ensure that demand for service $x \in X$ in segment $z \in Z$ is only allocated to $f \in F$ and $t \in T$ if f is active and both f and t are authorized to provide service x. Constraints (9) ensure that inter-facility flows of goods departing from f on network tier t_f can only be directed towards facilities on network tier $t_g = t_f + 1$. Here, M is an sufficiently large number. These constraints limit inter-facility flows to those between hubs and SFs and between SFs and LTPs. Constraints (10) link the decision variables ϕ^x_{fgv} and ψ^x_{fgv} and ensure that the number of vehicles deployed for the inter-facility transport is sufficient to transport the related parcel flow with ξ_v being the volumetric carrying capacity of vehicle type $v \in V$. Constraints (11) ensure that the capacity of facility f, ω_f, is higher than the sum of outgoing parcels at that facility. Constraints (12) ensure the flow conservation between incoming and outgoing parcels at SFs and LTPs. Constraints (13)–(16) define the domains of the decision variables.

4.2.2. Routing cost approximation

Problem-specific routing cost estimation In this section, we formulate the generalized cost for providing delivery service x in segment z from facility f through transportation mode t, C^x_{zft}. The RCE formulae presented in the following extends the route cost approximations presented by Smilowitz and Daganzo (2007), Winkenbach et al. (2016a), and Janjevic et al. (2019). As previously noted, our methodological contribution entails extensions to account for differentiated delivery services and alternative transportation modes, at a higher level of granularity. To this end, we introduce some additional notation that is summarized in Table S.1 in the Supplementary Material.

In essence, CA-based RCE methods seek to estimate the expected distance D, and more generally the expected cost, for a near-optimal tour to visit N customer locations over an area A. These methods build on the seminal work by Beardwood, Halton, and Hammersley (1959), who introduced the theorem $\lim_{N \to \infty} [D(N, A)] = \kappa \sqrt{NA}$, with κ being a constant that depends upon the distance metric used. In quadratic regions, upper bounds for κ have been introduced: $\kappa_{L_2} \le 0.90$ for the Euclidean metric and $\kappa_{L_1} \le 1.15$ for the rectilinear metric Daganzo (1984b). These approximation approaches build on the following assumptions: (1) The spatial distribution of PODs is governed by an homogeneous two-dimensional point process (Daganzo, 2005), and (2) the region is fairly compact and convex (Larson & Odoni, 2007). Several extensions to these original RCE methods have been studied that consider, for instance, the area size, the number of customer locations, multiple routes with capacitated vehicles, and time-windows (see, e.g., Ansari et al., 2018; Franceschetti, Jabali, and Laporte, 2017, and references therein). More recently, extensions to better account for real-world properties of the road-network and the impact of integrating pickups and deliveries on joint routes have been examined by Merchán and Winkenbach (2019) and Bergmann, Wagner, and Winkenbach (2020), respectively.

As stated previously, the work presented in this paper considers three types of last-mile delivery routes $r \in R$, with c, h and b representing *CDP routes*, *home delivery routes*, and *blended routes*, respectively. For any segment $z \in Z$, service type $x \in X$, transportation mode $t \in T$, and serving logistics facility $f \in F$, one or several route types can be deployed. Consequently, the generalized cost C^x_{zft} is equivalent to sum of the cost of individual route types,

$$C^x_{zft} = \sum_{r \in R} c^{xr}_{zft}, \qquad \forall z \in Z, f \in F, t \in T, x \in X. \tag{17}$$

For each route type $r \in R$, the operational characteristics of the route and the resulting cost c^{xr}_{zft} depend on a number of elements: (i) the presence of CDPs in a zone and the number of CDP parcels, (ii) the route type configuration for *home delivery routes*, and (iii) the possibility of serving CDPs jointly with home deliveries in *multi-stop tours*. Eqs. (18)–(25) provide the formulation of the specific cost per route type c^{xr}_{zft} that simultaneously accounts for these elements.

$$\begin{aligned} c^{xr}_{zft} = \mu^{xr}_{zft}\Big(& c^{\mathrm{km}}_t\big(2r_{zf} + (\zeta^{xr}_{zft} + \beta^r_z)\delta^{xr}_{zt}\big) \\ & + c^{\mathrm{H}}_t\Big(t^{\mathrm{l}}_f + \frac{2r_{zf}}{s^{\mathrm{l}}_t} + (\zeta^{xr}_{zft} + \beta^r_z)\frac{\delta^{xr}_{zt}}{s^{\mathrm{l}}_t} \\ & + \zeta^{xr}_{zft}(t^{\mathrm{d}}_t + \rho^x t^{\mathrm{i}}_t) + \beta^r_z(t^{\mathrm{d}}_t + \alpha^{xr}_{zt} t^{\mathrm{i}}_t)\Big) + c^{\mathrm{F}}_t \frac{1}{m^{xr}_{zft}} \\ & + c^{\mathrm{p}}_t\big(\zeta^{xr}_{zft}\rho^x + \alpha^{xr}_{zt}\big)\Big), \quad \forall z \in Z, f \in F, t \in T, x \in X, r \in R, \end{aligned} \tag{18}$$

where

$$\beta^{rx}_z = \begin{cases} 0, & \text{for } r = h, \\ min[0, \pi^x_z], & \text{for } r \in \{c, b\}, \end{cases} \qquad \forall z \in Z, x \in X, \tag{19}$$

$$\alpha^{xr}_{zt} = \begin{cases} 0, & \text{for } r = h, \\ \dfrac{\xi_t}{\theta^x} & \text{for } r = c, \\ \tau^x\Big(\pi^x_z - \lfloor \dfrac{\pi^x_z \theta^x}{\xi_t} \rfloor \dfrac{\xi_t}{\theta^x}\Big), & \text{for } r = b, \end{cases} \qquad \forall z \in Z, t \in T, x \in X, \tag{20}$$

$$\xi^{\hat{x}r}_{zft} = \begin{cases} \xi_t, & \text{for } r = h, \\ \xi_t - \dfrac{\xi_t}{\theta^x}, & \text{for } r = b, \end{cases} \qquad \forall z \in Z, f \in F, t \in T, x \in X, \tag{21}$$

$$T^{\hat{x}r}_{zft} = \begin{cases} T^m_{fx}, & \text{for } r = h, \\ T^m_{fx} - \beta^r_z(t^{\mathrm{d}}_t + \alpha^{xr}_{zt} t^{\mathrm{i}}_t), & \text{for } r = b, \end{cases} \qquad \forall z \in Z, f \in F, t \in T, x \in X, \tag{22}$$

$$\delta^{xr}_{zt} = \begin{cases} 0, & \text{for } r = c, \\ \lambda^x\Big[\dfrac{\kappa_z \kappa_t}{\sqrt{\dfrac{n^x_z + \tau^x \beta^{rx}_z}{A_z}}}\Big], & \text{for } r \in \{h, b\}, \end{cases} \qquad \forall z \in Z, f \in F, t \in T, x \in X, \tag{23}$$

$$\zeta^{xr}_{zft} = \begin{cases} 0, & \text{for } r = c, \\ (1 - \lambda^x) + \lambda^x min\Big[\dfrac{\xi^{\hat{x}r}_{zft}}{\rho^x \theta^x}, \dfrac{T^{\hat{x}r}_{zft} - t^{\mathrm{l}}_f - \frac{2r_{zf}}{s^{\mathrm{l}}_t}}{\frac{\delta^{xr}_{zt}}{s^{\mathrm{l}}_t} + t^{\mathrm{d}}_t + \rho^x t^{\mathrm{i}}_t}\Big], & \text{for } r \in \{h, b\}, \end{cases}$$

$$\forall z \in Z, f \in F, t \in T, x \in X, \tag{24}$$

$$m^{xr}_{zft} = max\left[1, \frac{T^{\mathrm{m}}_{fx}}{t^{\mathrm{l}}_f + \frac{2r_{zf}}{s^{\mathrm{l}}_t} + (\zeta^{xr}_{zft} + \beta^r_z)\frac{\delta^{xr}_{zt}}{s^{\mathrm{l}}_t} + \zeta^{xr}_{zft}(t^{\mathrm{d}}_t + \rho^x t^{\mathrm{i}}_t) + \beta^{rx}_z(t^{\mathrm{d}}_t + \alpha^{xr}_{zt} t^{\mathrm{i}}_t)}\right],$$

$$\forall z \in Z, f \in F, t \in T, x \in X, r \in R, \tag{25}$$

$$\mu^{xr}_{zft} = \begin{cases} \lfloor \frac{\pi^x_z \theta^x}{\xi_t} \rfloor + (1 - \tau^x)\Big(\frac{\pi^x_z \theta^x}{\zeta_t} - \lfloor \frac{\pi^x_z \theta^x}{\xi_t} \rfloor\Big), & \text{for } r = c, \\ \beta^{rx}_z \tau^x, & \text{for } r = b, \\ max[0, \frac{n^x_z - \mu^{xb}_{zft}\zeta^{xb}_{zft}}{\zeta^{xr}_{zft}}], & \text{for } r = h, \end{cases}$$

$$\forall z \in Z, f \in F, t \in T, x \in X. \tag{26}$$

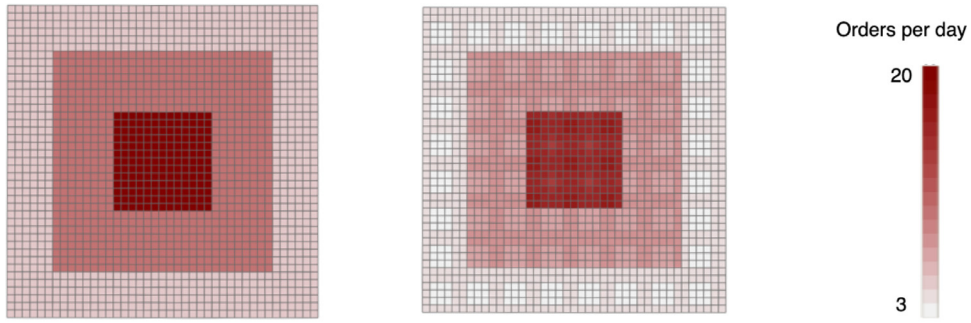
In Eq. (18), the cost c^{xr}_{zft} is obtained by multiplying the number of routes required to serve segment z, μ^{xr}_{zft}, by the cost of each individual route, represented by the term in parentheses. The cost of each individual route can be decomposed into four components.

The first route cost component (first line of term in parentheses) reflects the distance-based cost of a route. It is obtained by multiplying the distance-based operational cost of transportation mode t, c^{km}_t, by the total route distance. This distance is composed of (i) line-haul travel to and from the segment, $2r_{zf}$, and (ii) local inter-stop travel within the segment. The latter is the product of the expected inter-stop distance, δ^{xr}_z, and the average number of stops per route (i.e., home delivery stops, ζ^{xr}_{zft}, and CDP stops, β^r_z).

The second route cost component (second and third line of the term in parentheses) is a time-based cost of a route. It is obtained by multiplying the time-based operational cost of the transportation mode t, c^{H}_t, by the total route duration. This duration includes: (i) route setup duration t^{l}_f, (ii) line-haul travel duration $\frac{2r_{zf}}{s^{\mathrm{l}}_t}$, (iii) service zone travel duration (i.e. product of the average inter-stop travel duration, $\frac{\delta^{xr}_{zt}}{s^{\mathrm{l}}_t}$ by the average number of stops per route), (iv) service time duration for home delivery stops (product of average number of home delivery stops ζ^{xr}_{zft} by the service time duration composed of a fixed part t^{d}_t and a variable part $\rho^x t^{\mathrm{i}}_t$), and (v) service time duration of CDP deliveries (product of number of CDP stops, β^r_z by service time duration at a CDP, $t^{\mathrm{d}}_t + \alpha^{xr}_{zt} t^{\mathrm{i}}_t$, where α^{xr}_{zt} represents the number of parcels per CDP).

The third route cost component (fourth line of the term in parentheses) represents the fixed vehicle cost per route. It is obtained by dividing the daily vehicle fixed cost c^f_t of transportation mode t by the number of daily routes performed by each vehicle, m^{xr}_{zft}. The fourth route cost component (fifth line of the term in parentheses) represents the total parcel-based cost. It is obtained by multiplying the per-parcel delivery cost c^{p}_t of transportation mode t with the total number of parcels delivered per route $\zeta^{xr}_{zft}\rho^x + \alpha^{xr}_{zt}$.

Eq. (19) provides β^r_z, the number of CDP stops per route in segment z for route type r. Eq. (20) provides α^{xr}_{zt}, the number of CDP parcels per route for route type r. In addition, to establish the number of home delivery stops that can be performed on *home delivery* and *blended routes*, we consider two limiting factors: (i) vehicle capacity, and (ii) global service time. For *home delivery routes*, the entire vehicle capacity and the full global service time can be exploited to perform home delivery stops. For *blended routes*, a portion of the vehicle capacity is occupied by CDP parcels and a portion of the global service time is spent on CDP parcel delivery. To account for this, we introduce two additional terms: the *available* vehicle capacity for home-delivery stops, $\xi^{\hat{x}r}_{zft}$, and the *available* service time for home delivery stops, $T^{\hat{x}r}_{zft}$. The *available* vehicle capacity for home deliveries is provided in Eq. (21). For *home delivery routes*, it is equivalent to the initial vehicle capacity. For *blended routes*, it is obtained by subtracting the volume occupied



(a) Scenarios without CDPs (b) Scenarios with CDPs

Fig. 3. Density of home deliveries.

by CDP parcels delivered through the blended route, $\alpha_{zt}^{xr}\theta^{x}$, from the initial vehicle capacity, ξ_t. The *available* global service time for home deliveries is provided in Eq. (22). For *home delivery routes*, it is equal the maximum service time T_{fx}^{m}. For *blended routes*, we subtract the time required to service CDP deliveries, $\beta_z^r(t_t^{\mathrm{d}} + \alpha_{zt}^{xr} t_t^{\mathrm{i}})$, from the initial global service time, T_{fx}^{m}.

Apart from differences in $\xi_{zft}^{\hat{x}r}$ and $T_{zft}^{\hat{x}r}$, *home delivery* and *blended routes* are multi-stop tours with some common characteristics. For example, the average inter-stop travel distance between, δ_z^{xr}, is function of stop density and independent of vehicle capacity or global service time. Eq. (23) provides the generalized value of δ_z^{xr}, which is obtained in accordance with other CA-based RCE approaches stipulating that the average inter-stop distance is inversely proportional to the square root of the stop density (Beardwood et al., 1959; Daganzo, 1984a). Stop density calculations in a given segment z include home delivery stops, n_z^{x}, and, if available, CDP stops β_z^{rx}.

Eq. (24) provides ζ_{zft}^{xr}, the number of home delivery stops per route for *blended routes* and *home delivery routes*. We rely on the ARCE formulation by Winkenbach et al. (2016a) which considers two constraining factors to ζ_{zft}^{xr}, (i) the *available* vehicle capacity, and (ii) the *available* service time. The value of ζ_{zft}^{xr} is given by the minimum of value obtained when imposing these to constraining factors individually. Eq. (25) provides the number of routes that can be performed per day by a single vehicle, m_{zft}^{xr}. Finally, Eq. (26) provides the number of routes required for serving demand in segment z for delivery service x from facility f using transportation mode t and route type r, μ_{zft}^{xr}. To establish μ_{zft}^{xr} for *home delivery routes*, we divide the number of customers not served through *blended routes* by the number of customers that can be served per route, ζ_{zft}^{xr}.

Quality of approximation The critical model component driving the quality of our results is the CA-based RCE approach presented in this section. Although no academic benchmark instances consider similar problem sizes (i.e., several thousands of demand points), the quality of the approximation has been demonstrated on smaller-scale instances. First, as mentioned in our literature review, Winkenbach et al. (2016a) show that the approximation formulae produce solution values within a 10% range of those provided by tabu-search metaheuristics. Further, Janjevic et al. (2019) conduct an extensive benchmark analysis, comparing their approximate solutions to the best found explicit solutions to benchmark instances for the capacitated vehicle routing problem (CVRP) presented by Uchoa et al. (2017). Their results show that for realistic problem instance sizes, the estimated routing cost from a CA approach typically fall within 1 and 6% of the best-found exact solution. Similarly, Merchán and Winkenbach (2019) compare the quality of CA-based route distance estimates against near-optimal solutions obtained from a large neighborhood search heuristic, based on real road-network distances. Their results suggest that the mean absolute percentage error (MAPE) for these CA-based estimates ranges between 5 and 15%, depending on the connectivity of the underlying road network and the density of demand locations.

Janjevic et al. (2019) further evaluate the solution quality of a standard LRP that uses their CA-based RCE formula to replace the explicit routing component of the problem. They compare their CA-based solutions to the LRP instances presented by Schneider and Löffler (2019) with the best-known exact solutions to these problem instances. Despite the fact that these benchmark instances are still substantially smaller than the realistically sized instances we are considering in this work, their solutions to these instances on average only deviate by around 3% from the best-known exact solution.

5. Stylized experiments

To illustrate the use of our proposed model, we conduct a number of stylized numerical experiments using synthetic data. We investigate the resulting network configuration based on multiple demand scenarios, inspired by the real-life case study of a distribution network of a large e-commerce retailer in São Paulo, Brazil, which we discuss further in Section 6. The operational and cost parameters relevant to demand segments, transportation modes, and logistics facilities used in these experiments are inspired by actual data provided by the retailer.

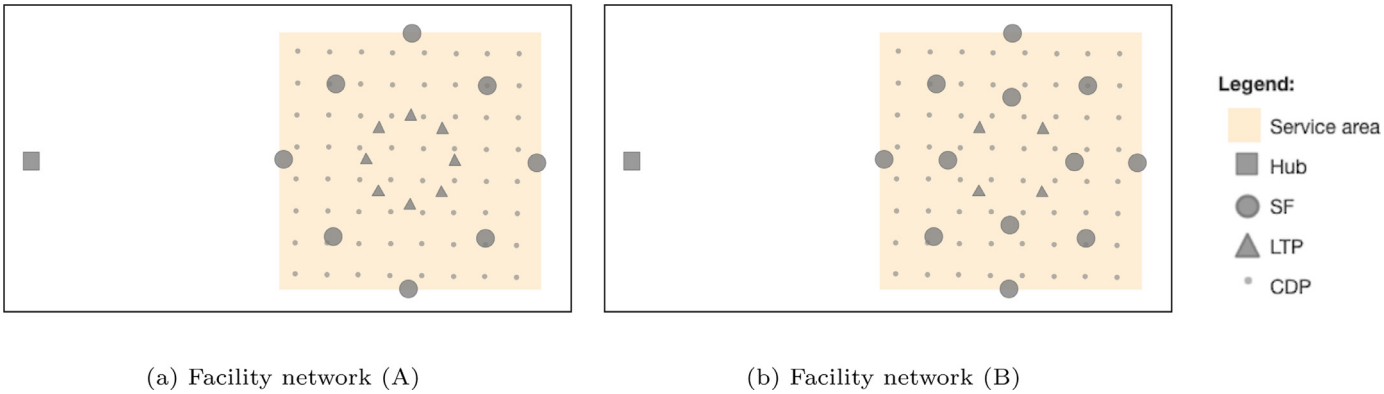
5.1. Description of numerical experiments

In the following, we present the main parameters characterizing our numerical experiments. The complete set of instances is provided as an electronic appendix to this paper.

Customer demand We consider deliveries in a rectangular service area of 1,681 km^2 with a total demand of 14,300 orders corresponding to 15,730 parcels per day. The service area is subdivided into square-kilometer segments arranged in a 41x41 grid. Each segment belongs to one of three sub-areas of low (5 orders / day), medium (10 orders / day), and high demand density (20 orders / day), as depicted in Fig. 3. Depending on the demand scenario considered, deliveries either exclusively occur at individual customer homes or at a combination of individual customer homes and CDP locations. Based on findings from the literature (Morganti et al., 2014; UPS, 2015, see, e.g.,), we assume that 20% of total demand can be captured by CDPs. We assume that for any given segment, the fraction of demand that is attracted by a given CDP decreases linearly in the distance between the segment centroid and the CDP location. Thus, the density of home deliveries in a segment is reduced more strongly if the segment is closer to a CDP (see Fig. 3).

Table 4
Percentage of parcels per delivery service for each demand scenario.

		STD (home)	STD (CDP)	EXP	INST	Total
NoDiv	Baseline scenario (standard home deliveries)	100	0	0	0	100
NoDiv-CDP	Baseline with home and CDP deliveries	80	20	0	0	100
MedDiv	Medium service diversification	85	0	10	5	100
MedDiv-CDP	Medium service diversification with CDPs	65	20	10	5	100
HiDiv	High service diversification	70	0	20	10	100
HiDiv-CDP	High service diversification with CDPs	50	20	20	10	100



(a) Facility network (A) (b) Facility network (B)

Fig. 4. Schematic representation of the types and locations of candidate logistics facilities considered.

Time-differentiated delivery services We consider three types of delivery services: (1) standard deliveries with a lead time of at least 1 day (denoted as "STD"), (2) express deliveries with a lead time of 6 hours (denoted as "EXP"), and (3) instant deliveries with a lead time of 4 hours (denoted as "INST"). For standard deliveries, both homes and CDPs are served. The other two service types are only available for home deliveries. For standard deliveries, order preparation at the hub and transfer from the hub towards downstream facilities in the network are assumed to occur on a day before the delivery. We therefore consider a global service time of 8 hours to be available to complete these deliveries whether they originate from SFs or LTPs. For express and instant deliveries, order preparation and inter-facility transfers occur on the day of the delivery. Therefore, the available global service times for these services are decreased by one hour for SFs and two hours for LTPs. Based on these assumptions, we investigate multiple demand scenarios with varying delivery service mix, summarized in Table 4.

Logistics facilities and inter-facility transportation In the subsequent analyses we consider two generic networks of potential distribution facilities. The locations of available logistics facilities in both networks are shown in Fig. 4. Both networks assume a single hub with a capacity of 20,000 parcels / day. Facility network (A) assumes 8 SF candidates and 8 LTP candidates. Facility network (B) assumes 12 SF candidates and 4 LTP candidates. SFs and LTPs have a capacity of 6000 and 600 parcels / day in both networks. The fixed cost of the hub is set to 4000 cost units per day. The fixed cost of SFs varies between 2400 and 3600 units per day. Here, to capture differences in real-estate cost, more centrally-located SFs are assumed to incur a higher cost. The fixed cost of LTPs is set to 1,200 units per day. Inter-facility transportation links can be established between the hub and the SFs, as well as between the SFs and the LTPs. Inter-facility transport is performed by specialized vans ($c_v^{\text{km}} = 5,\ \xi_v = 10$).

Last-mile delivery routes. We consider multiple last-mile transportation modes that differ according to the type of vehicle employed and the governance of fleet operations: (i) minivans (company-owned fleet), (ii) minivans (outsourced), (iii) cargo-bikes (company-owned fleet), (iv) motorbikes (outsourced), and (v) passenger cars (crowd-shipping). All transportation modes can be employed across all delivery services with the exception of instant deliveries, which can only be served through outsourced transportation modes and crowd-shipping. Cost and operational parameters are provided in Table 5.

5.2. Results

In the following, we summarize the results from our numerical analyses[1] to assess (1) the impact of accounting for differentiated delivery services during the network design on its optimal configuration and cost, (2) the impact of accounting for granular network features during the network design on its optimal configuration and cost, and (3) the benefits of an integrated optimization approach to designing the distribution network. Please note that there are extensive illustrations of our detailed analysis results available as a Supplementary Material to this paper.

5.2.1. Value of accounting for differentiated delivery services

To investigate how accounting for differentiated delivery services during the network design impact the optimal network configuration and cost performance, we analyze our numerical results for the six demand scenarios presented in Table 4 for facility network (A).

Network design. Table 6 summarizes the obtained optimal design of the omni-channel distribution network for the various demand scenarios we consider. Specifically, it shows the fraction of demand for every service type that is addressed from the various facility types in the network, as well as the number of active facilities of each type that are handling these package volumes. Fig. 5 illustrates the facility configurations and inter-facility flows in the optimized networks for all six demand scenarios. Figs. S.1–S.6 in the Supplementary Material provide a more detailed view on the obtained facility configurations and service area allocations. We can make two main observations. First, when comparing the facility configuration in demand scenarios *NoDiv*, *MedDiv* and *HiDiv*,

[1] Implemented in Python 3, using Gurobi 6.5.1 as numerical solver. Executed on 2.7 GHz Intel Core i7 chipset, 16 GB of memory, OS X Mojave. MIP optimality gap set to 0.2%. See Table S.2 in the Supplementary Material for computation times.

Table 5
Cost and operational parameters of last-mile transportation modes.

Last-mile transp. mode	Cost parameters				Operational parameters						
	c_t^{km}	c_t^{p}	c_t^{H}	c_t^{F}	t_t^{l}	κ_t	s_t^{l}	s_t^{s}	t_t^{d}	t_t^{i}	ξ_t
minivans (company-owned fleet)	0.4	0	20	70	0.5	0.8	40	20	0.05	0.01	2.5
minivans (outsourced)	0	0	35	0	0.5	0.8	40	20	0.05	0.01	2.5
cargo-bikes (company-owned fleet)	0.02	0	15	10	0.3	0.7	7	7	0.03	0.01	1
motorbikes (outsourced)	0	0	25	0	0.2	0.7	40	20	0.04	0.01	0.03
passenger cars (crowd-shipping)	0.4	5	0	0	0.1	0.8	40	20	0.04	0.01	0.3

Table 6
Share of parcels handled by each facility type per delivery service type and scenario.

scenario	service	% via LTPs (# LTPs)	% via Central SFs (# SFs)	% via Peripheral SFs (# SFs)
NoDiv	STD	– (0)	– (0)	100.0 (3)
NoDiv-CDP	STD	– (0)	67.0 (2)	33.0 (1)
MedDiv	STD	9.0 (3)	91.0 (3)	– (0)
	EXP	– (0)	100.0 (3)	– (0)
	INST	57.0 (3)	43.0 (3)	– (0)
	total	*10.5 (3)*	*89.5 (3)*	*– (0)*
MedDiv-CDP	STD	6.2 (2)	58.9 (2)	34.9 (1)
	EXP	20.8 (1)	59.3 (2)	19.9 (1)
	INST	63.6 (3)	29.1 (2)	7.3 (1)
	total	*10.5 (3)*	*57.5 (2)*	*32.0 (1)*
HiDiv	STD	12.9 (4)	87.1 (4)	– (0)
	EXP	– (0)	100.0 (4)	– (0)
	INST	49.8 (4)	50.2 (4)	– (0)
	total	*14.0 (4)*	*86.0 (4)*	*– (0)*
HiDiv-CDP	STD	12.9 (4)	87.1 (4)	– (0)
	EXP	– (0)	100.0 (4)	– (0)
	INST	49.8 (4)	50.2 (4)	– (0)
	total	*14.0 (4)*	*86.0 (4)*	*– (0)*

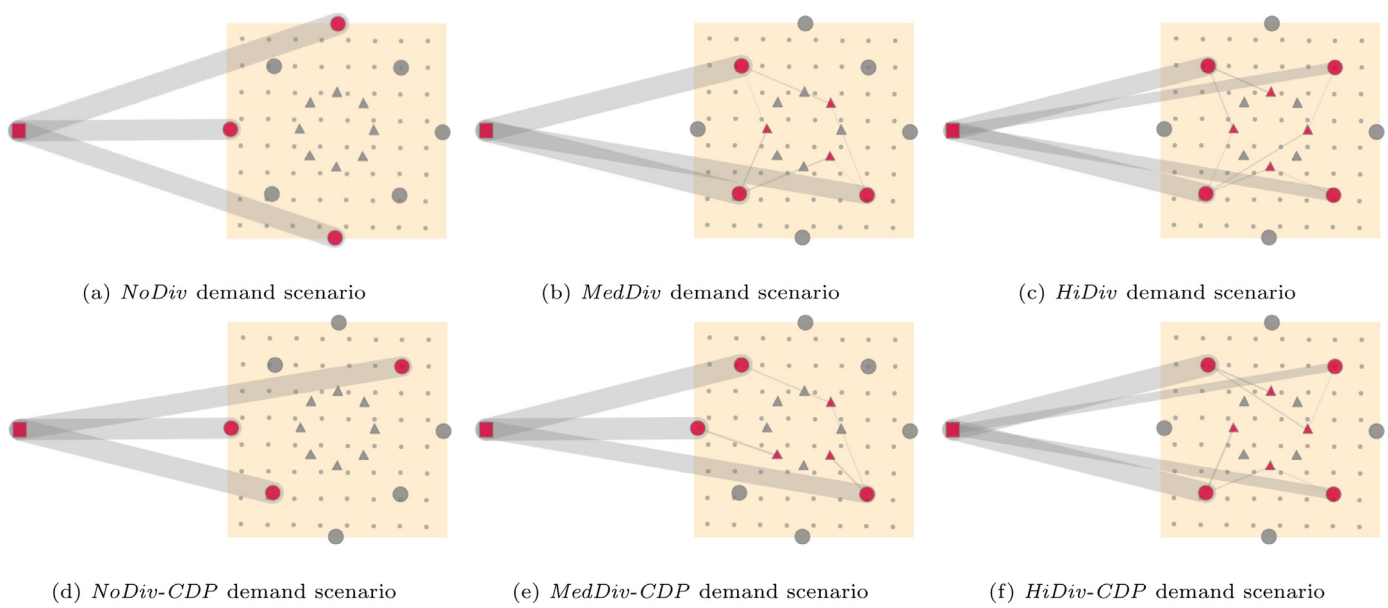

(a) *NoDiv* demand scenario
(b) *MedDiv* demand scenario
(c) *HiDiv* demand scenario
(d) *NoDiv-CDP* demand scenario
(e) *MedDiv-CDP* demand scenario
(f) *HiDiv-CDP* demand scenario

Fig. 5. Network configurations obtained.

we observe that increasing the share of express and instant deliveries leads to an increase in the number of facilities deployed and promotes the use of facilities in closer proximity to high demand areas. Second, when comparing the network configurations in the *NoDiv*, *MedDiv* and *HiDiv* demand scenarios with their counterparts with CDPs (*NoDiv-CDP*, *MedDiv-CDP*, and *HiDiv-CDP*), we observe that the introduction of CDPs changes the optimal facility configuration and/or the inter-facility flows across all three scenarios.

Cost performance. Table 7 shows the average cost per parcel for different demand scenarios, broken down by cost components and delivery service types. Our results are intuitive. First, we observe that a decrease in the delivery lead time (i.e., a stronger weight of faster delivery services) increases the cost per parcel. The main factor driving this increase in total cost is the cost of last-mile distribution. Second, our results show that the introduction of CDPs decreases the total cost per parcel. The underlying rationale is that CDPs aggregate demand in fewer points of delivery, thus reducing the cost of last-mile distribution. Third, our experiments confirm that as the relative demand share of specific delivery service increases, the cost per parcel shipped through that particular service decreases. Since the density of demand served by a certain delivery service increases with its relative share, a higher relative share allows for more efficient last-mile operations.

Table 7
Average cost per parcel per delivery service type and scenario (normalized).

Scenario	Service	Fixed facility	Inter-facility transfer	Inter-facility transport	Handling	Last-mile distribution	Total	Diff.[a]
NoDiv	STD	0.71	0.44	1.05	0.26	3.88	6.34	–
NoDiv-CDP	STD	0.86	0.44	0.99	0.26	3.26	5.81	−8.4
MedDiv	STD	1.15	0.44	1.09	0.26	3.62	6.56	+3.5
	EXP	1.02	0.44	1.2	0.26	7.37	10.28	+62.2
	INST	1.83	0.45	2.57	0.26	14.72	19.83	+213.0
	avg.	*1.17*	*0.26*	*4.55*	*0.44*	*1.17*	*7.59*	*+19.8*
MedDiv-CDP	STD	1.02	0.44	0.95	0.26	3.18	5.86	−7.6
	EXP	1.28	0.44	1.26	0.26	7.19	10.44	+64.7
	INST	1.91	0.45	2.45	0.26	15.5	20.57	+224.6
	avg.	*1.09*	*0.26*	*4.20*	*0.44*	*1.06*	*7.05*	*+11.3*
HiDiv	STD	1.46	0.44	1.15	0.26	3.51	6.82	+7.7
	EXP	1.32	0.44	1.17	0.26	6.08	9.26	+46.2
	INST	1.9	0.45	1.82	0.26	12.86	17.28	+172.7
	avg.	*1.47*	*0.26*	*4.96*	*0.44*	*1.22*	*8.36*	*+31.9*
HiDiv-CDP	STD	1.43	0.44	1.1	0.26	2.8	6.04	−4.8
	EXP	1.38	0.44	1.17	0.26	6.08	9.32	+47.1
	INST	1.95	0.45	1.82	0.26	12.86	17.33	+173.4
	avg.	*1.47*	*0.26*	*4.46*	*0.44*	*1.18*	*7.82*	*+23.4*

[a] difference in percent compared to cost of standard deliveries in *NoDiv* demand scenario.

Table 8
Share of transportation modes deployed according to the demand scenario and delivery service (percent of parcels).

		Minivan (Fleet)	Cargo-bike (Fleet)	Minivan (Outsourced)	Motorbike (Outsourced)	Crowd-shipping
NoDiv	STD	84.4	6.8	8.6	–	0.2
NoDiv-CDP	STD	81.5	9.5	9.0	–	–
MedDiv	STD	67.9	7.3	24.8	–	–
	EXP	–	1.9	45.9	–	52.2
	INST	–	–	–	45.1	54.9
MedDiv-CDP	STD	76.1	14.2	9.8	–	–
	EXP	–	2.5	41.3	–	56.2
	INST	–	–	–	40.9	59.1
HiDiv	STD	66.4	33.6	22.5	–	–
	EXP	–	5.9	84.5	–	9.6
	INST	–	–	–	58.6	41.4
HiDiv-CDP	STD	75.7	24.2	0.1	–	–
	EXP	–	5.9	84.5	–	9.6
	INST	–	–	–	58.6	41.4

Transportation modes. Table 8 summarizes relative mix of transportation modes deployed across the various delivery services for all demand scenarios considered. We observe that the delivery service type has a major impact on the transportation mode deployed. For example, cargo-bikes, characterized by a relatively low speed, are mostly deployed for standard deliveries and appear less attractive for shorter delivery lead times. Motorbikes, in turn, due to their limited capacity, are only an attractive option for very short lead times, i.e., instant deliveries. We further note that delivery services with shorter delivery lead times generally favor outsourcing of transportation. While across all demand scenarios, over 90% of standard deliveries are served by a company-owned vehicle (minivans or cargo-bikes), this number is less than 6% for express deliveries.

Fig. 6 provides a geo-spatial mapping of where the various available transportation modes are deployed for a subset of scenarios and delivery services. The complete mapping of all demand scenarios and delivery services is provided in Figs. S.1–S.6 in the Supplementary Material. We observe that the distance between a segment and a facility impacts transportation modes deployed. For example, in case of standard deliveries, the choice of transportation mode in a segment changes from cargo-bikes to minivans and outsourced minivans as the distance to the facility increases. In case of instant deliveries, motor-bikes are generally deployed in the proximity of logistics facilities whereas crowd-shipping is generally deployed in more distant areas. For express deliveries, cargo-bikes are predominantly deployed close to logistics facilities.

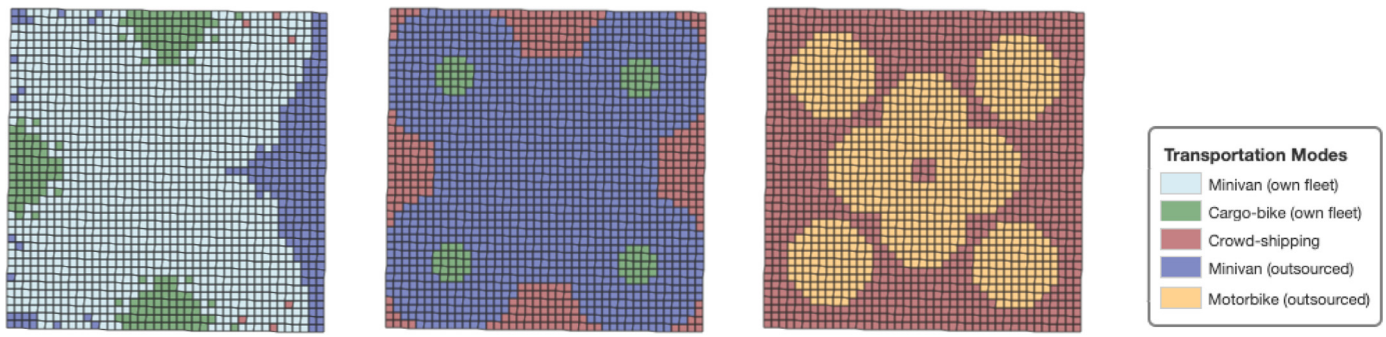
We further observe that besides the effect of distance, the choice between company-owned and crowd-shipped or outsourced transportation is driven by demand density in a segment. For instance, express deliveries are served through crowd-shipped deliveries except for areas that either have a very high demand density or have a medium demand density and are located close to the facility.

5.2.2. Value of accounting for granular network features

To investigate the impact of accounting for the granular features of last-mile distribution networks for omni-channel retailers (i.e., third echelon of facilities, and a variety of competing last-mile transportation modes), we compare how the optimal network configuration and cost performance changes as some of these features are not considered during the network design. We present our main results in the following.

Three echelons of distribution. We investigate the impact of the introduction of the third echelon of facilities by comparing results of our model with a *two-echelon* network design model (i.e., setting with hubs and SFs only). We observe no change in the *No-Div* scenario, as only SF candidates are deployed. On the other hand, in the *HiDiv* demand scenario, the *two-echelon* model leads to a cost increase of 3.8%, mainly driven by increase in last-mile distribution cost (14.0%). Instant deliveries undergo the highest increase in cost (25.0%), confirming that three-echelon setting with hyper-local facilities is a cost-effective option for fast delivery services.

Heterogeneous transportation modes and configurations of last-mile delivery routes. We investigate how the variety of last-mile transportation modes employed affects network performance. We first consider *NoDiv* scenario and restrict transportation modes to company-owned fleet minivans and achieve an increase in cost of



(a) *NoDiv*, STD deliveries (b) *HiDiv*, EXP deliveries (c) *HiDiv*, INST deliveries

Fig. 6. Examples of transportation modes deployed.

1.3%. We then consider *HiDiv* demand scenario and compare costs of the original setting with costs in a setting with limited transportation modes (i.e., own-fleet and outsourced minivans only). We find that restricting available transportation modes increases costs by 11.9%. These results confirm that the diversification of the transportation modes decreases distribution costs and that fast-delivery services are particularly sensitive to the transportation mode mix.

5.2.3. Value of an integrated optimization approach

In this section, we assess the advantages of our proposed *integrated optimization approach*, described in the previous sections over other approaches that do not comprehensively integrate all the decisions inherent to the problem.

Accounting for three tiers of facility location decisions during network design. We analyze how the integration of facility location decisions on a second and third echelon impacts the performance of the resulting distribution network. For *HiDiv* demand scenario, we compare results of our integrated model with a *hierarchical* approach, where we first establish the SFs to activate based on a two-echelon model, and then introduce LTPs. For the available facility network (A) (see Fig. 4 a)), both approaches yield identical network designs, given the limited number of SF candidates. As we provide additional flexibility in choosing the optimal facility locations to the model, assuming available facility network (B), we observe that our integrated design approach produces the same network configuration as previously, while the *hierarchical* approach yields a different configuration. The cost of the design found by the *hierarchical* approach is 2.2% higher, confirming that the integration of facility location decisions on two echelons leads to more cost-effective designs if design choices are sufficiently free (i.e., if enough relevant candidate locations are available to choose from). Detailed results of the integrated and the hierarchical network design model for the available facility network (B) are reported in Figs. S.7 and S.8 in the Supplementary Material.

Accounting for diversified delivery services during network design To assess the value of considering multiple available delivery services simultaneously while designing the last-mile distribution network, we focus on the *HiDiv* demand scenario and compare the results of our integrated model with those from a *sequential* approach that progressively introduces different delivery services. Specifically, in the sequential approach, all demand is assumed to be served by a single, dominant delivery service (here, standard delivery) when the strategic decisions on activating facility locations are made. These facility decisions are then fixed before introducing express and instant delivery services and re-optimizing the remaining (non-facility) decisions. Here we distinguish two cases: (1) *fixed facility configuration,* where we disallow the activation of additional facilities, and (2) *flexible facility configuration*, where we allow additional facilities to be activated in response to the newly introduced additional delivery services. Figures S.9 and S.10 in the Supplementary Material provide a geo-spatial representation of the results of both variants of the *sequential network design model*.

Compared to a *sequential* design approach with *fixed facility configuration* our *integrated network design model* leads to a reduction in total network cost by 11.6%. Even compared to a *sequential* design approach with *flexible facility configuration*, our integrated approach still yields a cost reduction by 2.9%. The superior cost performance of a distribution network resulting from our integrated design approach is predominantly driven by significant reductions in the obtained last-mile distribution cost. Specifically, our integrated approach reduces last-mile cost by 28.9% (13.7%) compared to a sequential design approach with fixed (flexible) facility configuration. The cost of instant deliveries are particularly affected by a sequential design approach. Compared to an integrated approach, they suffer from a 58.8% (15.2%) increase in the overall per-parcel cost in case of a fixed (flexible) facility configuration. Similar results can be observed for other demand scenarios as well.

The two results presented above confirm that network design approaches that assume a single, dominant delivery service when making strategic facility decisions, lead to sub-optimal designs and significant cost inefficiencies in other delivery services. While introducing flexibility in subsequently adjusting the facility network can mitigate the shortcomings of *sequential* design approaches to a certain extent, it fails to entirely close the performance gap in comparison with our *integrated* approach.

Accounting for multiple product exchange options during network design To explore how the integration of CDPs in the network design model impacts the cost of the solutions obtained, we compare results of our integrated approach with a non-integrated alternative. In the non-integrated case, we ignore the presence of CDPs when making strategic decisions about active facility locations. We then fix the obtained facility network and shift a portion of demand towards the CDPs. While we observe differences in the network configurations obtained from the integrated and the non-integrated approach, the observed cost differences between both configurations are typically less than 1%. We believe that the limited cost impact of integrating CDPs in the network design is due to our assumption of CDPs being uniformly distributed across the service area and being governed by the same demand attraction function for every CDP. Therefore, the introduction of CDPs leads to local shifts in demand, but only minor changes in high-level demand patterns. In practice, areas with high population and demand density are likely to be addressed with a higher number of more densely located CDPs (Morganti et al., 2014). In that case, higher density areas would also be more affected by the activation of CDPs, which may lead to a stronger effect of CDPs on the optimal network design and cost performance.

Accounting for competing transportation modes during network design To explore how the integration of multiple transportation modes in the network design impacts the cost and configuration

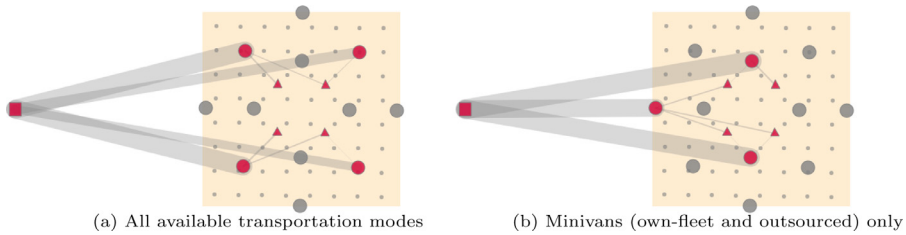


Fig. 7. Examples of facility configurations obtained for *Med-Div* scenario and facility network (B).

of the network obtained, we compare results of our integrated approach with a non-integrated alternative, where facility locations are established considering a subset of transportation modes and other modes are subsequently deployed. While we find differences in the network configuration obtained for some scenarios (see, e.g., Fig. 7), we find relatively small differences in the network cost (around 2% for the example on Fig. 7).

Further, we analyze how a diversified set of available transportation modes affects the value of considering (1) granular network features (i.e., a third echelon of distribution facilities, see Section 5.2.2) and (2) a diversified set of available delivery services (see Section 5.2.1) in an integrated network optimization approach. Focusing on the the *HiDiv* demand scenario for illustration, we can make two main observations. First, when restricting the available transportation modes to minivans (own-fleet and outsourced), the difference in cost between the two-echelon and the three-echelon network design model rises from 3.8% to 9.2%. Similarly, the cost difference between the *integrated* and *hierarchical* network design approach rises from 2.2% to 2.4%. This illustrates interrelations between the different dimensions of the decision problem. When a wider set of transportation modes is available, changes in the transportation mode mix can help to partially mitigate the shortcomings of a two-echelon and/or *hierarchical* network design approach approach.

Second, restricting the available transportation modes can lead to a sequential design approach with fixed facility configuration to suggest a network design that makes it infeasible to provide some of the delivery services with shorter lead times. This confirms that if the mix of delivery services is ignored in the network design phase, and in particular in the choice of logistical facilities, flexibility in transportation modes is required in order to still be able to fulfill all types of customer demand. For a sequential design approach with flexible facility configuration, the problem remains feasible even as the set of available transportation modes reduces. However, the cost differential between an integrated and a sequential design approach rises from 2.9% to 4.9%, as the sub-optimal initial network design needs to be expanded more significantly to cater for faster delivery services.

6. Case study

We apply the proposed model to a real-world case study inspired by the distribution network of a large e-commerce retailer in São Paulo, Brazil. The company serves approximately 15,000 daily orders across the study area and offers one standard delivery service, corresponding to a delivery service mix equivalent to the *NoDiv* scenario presented in Section 5. Fig. 8 shows the company's daily demand density in the São Paulo metropolitan area. Demand is aggregated in daily orders per square kilometer. Zones receiving fewer than 10 orders per year are omitted, leaving us with 2400 square-kilometer segments. The company employs a two-echelon distribution network with one distribution hub and three SFs. Last-mile distribution is provided using a combination of company-owned and outsourced minivans. In the following, we

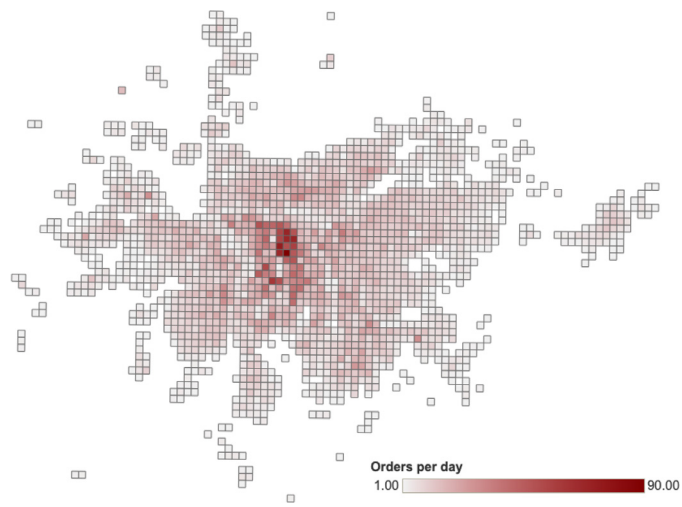


Fig. 8. Orders per day per square kilometer in São Paulo study area, from 2015 operational data shared by retailer.

refer to the current demand scenario with these current network features as *NoDiv (current)*.

Growing congestion and regulatory pressure in São Paulo, as well as changing customer preferences for faster delivery services are forcing the company to reconsider its last-mile network design. Specifically, it seeks to establish a third tier of logistics facilities and new last-mile transportation modes. For the redesign of its network, the company has considered seven SF candidates (capacity between 1500 and 6000 parcels / day) and six LTP candidate locations (capacity of 500 parcels / day). Further, the company seeks to explore if cargo-bikes could be a cost-effective option for serving demand in very dense areas, and if motorbikes and crowd-shipped deliveries can decrease overall delivery costs in specific zones of the city. In a first analysis (referred to as *NoDiv (new)*) we quantify the impact of allowing for these new network features on the optimal network configuration and cost performance given the current demand (see Tables 9 and 10).

While expedited delivery services only represent a negligible share of the company's total deliveries nowadays, the company projects high growth rates for these services. In addition, the company seeks to leverage its network of brick-and-mortar stores as CDPs for e-commerce deliveries. To study the impact of both, changes in delivery service preferences and integrating CDPs, we analyze additional demand scenarios, including the *HiDiv-CDP* scenario presented in Section 5 with both the *current* and *new* network features. We find that the *current* network features are unable to accommodate the demand for faster delivery services in the *HiDiv-CDP* scenario, confirming the necessity to align the company's distribution strategy and the delivery services it offers with the design of its network. We therefore only report results of the *HiDiv-CDP* scenario assuming the *new* network features in

Table 9
Average cost per parcel per delivery service type and scenario (normalized).

Scenario	Service	Fixed facility	Inter-facility transfer	Inter-facility transport	Handling	Last-mile distribution	Total	Diff.[a]
NoDiv (current)	STD	20.9	7.2	5.4	4.8	61.7	100.0	–
NoDiv (new)	STD	20.1	7.5	3.7	5.0	63.6	100.0	−4.1
HiDiv-CDP (new)	STD	23.1	8.2	3.9	5.4	59.3	100.0	−11.2
	EXP	15.4	5.2	3.0	3.4	72.9	100.0	+39.8
	INST	8.1	2.8	2.6	1.8	84.7	100.0	+172.0
	avg.	*17.8*	*6.3*	*3.4*	*4.1*	*68.4*	*100.2*	*+17.3*

[a] difference in percent compared to the standard deliveries in *NoDiv (current)* scenario.

Table 10
Share of transportation modes deployed according to the demand scenario and delivery service (percents).

		Minivan (own fleet)	Cargo-bike (own fleet)	Minivan (outsourced)	Motorbike (outsourced)	Crowd-shipping
NoDiv (current)	STD	78.6	–	21.4	–	–
NoDiv (new)	STD	56.5	20.5	22.7	–	0.3
HiDiv-CDP (new)	STD	46.2	18.7	34.3	–	0.8
	EXP	–	2.0	92.9	–	5.0
	INST	–	–	–	78.9	21.1

Tables 9 and 10. More detailed illustrations of these results are presented in Figures S.11 and S.12 in the Supplementary Material.

In the *NoDiv (new)* implementation, one LTP is activated in addition to three SFs, and a share of deliveries is shifted from company-owned minivans to cargo-bikes. This leads to a decrease of the overall cost by 4.1% compared to the status quo, i.e., the *NoDiv (current)* implementation. Comparing the *HiDiv (new)* and *NoDiv (new)* results reveals the same trends as in our stylized experiments in Section 5: A higher diversification of demand towards faster delivery services leads to increasing the number of active facilities and active facility locations moving closer to high demand areas. Also, we see crowd-shipping and motorbikes being used for instant deliveries. Similarly, we observe a cost increase for expedited delivery services, while the cost of standard deliveries are reduced due to the introduction of CDPs.

The results presented here are representative of a whole range of scenario analyses we conducted for the company and which informed its decisions associated to the restructuring of its urban last-mile distribution network and the roll-out of new delivery service offerings and transportation modes. For instance, the decision to introduce LTPs as a central element of the company's future distribution strategies was a result of these analyses. Similarly, the introduction of cargo-bikes as a potential future transportation mode was a result of our model-based analyses, which helped the company to assess whether a cost efficient future last-mile distribution footprint could be aligned with more sustainable transportation options. While the full implementation of our findings will require time and additional analyses based on our proposed model, initial investments into LTPs, CDP, and other elements of the company's revised urban distribution infrastructure have been made.

7. Discussion and managerial insights

The results from Sections 5 and 6 yield several relevant insights into trade-offs in last-mile distribution network design for omni-channel retailing. We summarize these in the following.

More granular network features offer opportunities to mitigate the cost impact of emerging last-mile delivery services To keep up with rising customer expectations, e-retailers are offering time-differentiated delivery services that come at the expense of lower delivery efficiency and higher cost. We demonstrate that the deployment of facilities in close proximity to the customer, such as LTPs, and the use of outsourced and crowd-sourced transportation modes can help e-retailers to enable these services, while partially mitigating associate cost effects. In addition, CDPs, while offering customers flexibility and convenience, also reduce transportation costs as they consolidate deliveries.

Distribution network design and transportation mix needs to be closely aligned with a company's delivery service offerings. Our results show that different delivery services with different lead times are optimally offered via different types of network facilities and transportation modes. In addition, we find that the presence of CDPs can impact both the optimal transportation mix and the configuration of logistics facilities.

Distribution strategies need be tailored to local service area characteristics. We show that several characteristics of the considered service area affect the optimal distribution strategy to be deployed. For instance, cargo-bikes and motorbikes are cost-effective to serve demand in close proximity to a facility. Further, the choice between in-sourced, outsourced, or crowd-sourced modes depends on the density of the demand zone and its proximity to the facility. This implies that a granular characterization of the service area is needed. By dividing the area into small rectangular segments, we were able to derive fine-grained cost estimates that ultimately improve the level of fidelity in our results. More aggregate units of spatial analysis (e.g., postal codes) would lead to a reduced data collection effort at the expense of a less granular characterization of demand, allowing for a less accurate depiction of the cost trade-offs between competing facility locations and transportation modes.

Fully integrating the various design decisions yields better-performing networks. The experiments conducted as part of our analysis show that an integrated network design approach, as proposed in this paper, systematically outperforms approaches that solve sub-problems of the overall network design problem in isolation. Failure to integrally consider the highly interrelated decisions on facility locations and types as well as transportation modes in a comprehensive model could therefore result in network configurations that are unable to cost-effectively support time-differentiated delivery service offerings with multiple product exchange options. In effect, this would put the success of any state-of-the-art omni-channel retailing strategy at risk. However, we also show that being able to choose from a more diverse set of facilities and from a wider range of diversified transportation modes can partially mitigate the shortcomings of non-integrated design approaches.

8. Conclusion

In this study, we propose a multi-echelon location-routing problem (ME-LRP) that captures several relevant features of the

operating environment of contemporary e-commerce last-mile distribution systems. We perform numerical analyses on several stylized experiments before applying our model to a real-world case study of urban last-mile distribution by a major Brazilian e-commerce platform.

Our results can be summarized in two main findings. First, we demonstrate the need for a *granular approach* for last-mile distribution network design in an omni-channel environment. Through multiple experiments based on stylized and real data, we show benefits of discriminating between different delivery services, product exchange options, facility types, and last-mile transportation modes when determining the optimal strategic design of the last-mile distribution network. For example, we demonstrate that the offered delivery service mix has a significant impact on the optimal network design decisions and that heterogeneous transportation modes are required when offering highly-responsive delivery services. These results are in line with current trends observed in e-commerce distribution, with companies progressively deploying hyper-local distribution networks and using a wider variety of delivery options with varying lead times.

Second, we confirm the need for an *integrated approach* to network design that considers all aspects of the problem jointly and simultaneously. For example, we demonstrate that non-integrated design approaches that sequentially or hierarchically optimize for subsets of the relevant design decisions produce network designs with inferior cost performance. We also show that the shortcomings of such non-integrated design approaches can in part be mitigated by enabling a posteriori flexibility to adjust the network configuration (e.g., by opening additional facilities, or altering the transportation mix).

A number of future extensions for this model may be suggested. For instance, our model does not currently allow for route consolidation across different delivery services, whereas operating delivery services with similar spatial and temporal demand characteristics on joint routes may yield cost synergies. Moreover, our model currently assumes that all inventory is held cenrtrally in the hub. Extending our model to allow for local inventory decisions at facilities across all tiers of the network would constitute a fruitful path for future research on omni-channel fulfillment strategies that leverage local store networks and other forms of localized stockholding and order preparation.

Further, we currently consider that the demand for all delivery services is given as an input to our analysis, and we focus on the minimization of distribution cost only. A useful extension may consist in adopting a revenue management perspective to determine a profit optimal configuration of the distribution network and involves and active decision of the retailer on where to offer which delivery service type based on local differences in profitability (cf., Snoeck, Merchán, & Winkenbach, 2020). Similarly, integrating the decision on where to deploy CDPs would allow for an active tradeoff between the cost of placing and operating a CDP and the reduction in the cost of deliveries it would enable. Letting the network design model actively choose the number and locations of CDPs would give rise to the need to explicitly model customer preferences and behavior. The attraction of customers to a CDPs would no longer be a fixed given. Rather, every CDP that is placed by the model would affect customer utility from using a particular delivery service or product exchange option. A change in the number an location of CDPs would thus result in a shift in demand distribution across delivery services and product exchange options.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2020.08.043.

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